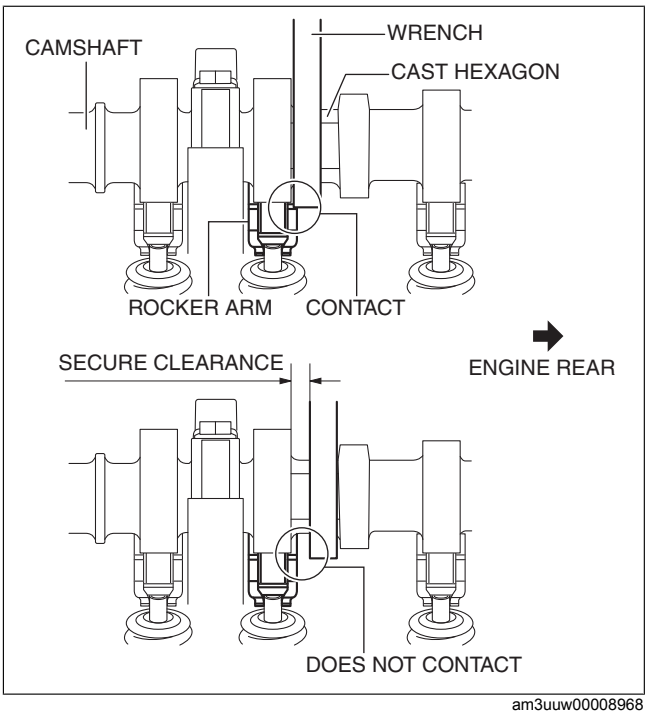
## TIMING CHAIN ASSEMBLY

id011000505600

**Caution**

- **If the camshaft is rotated with the timing chain removed and the piston at the top dead center position, the valve may contact the piston and the engine could be damaged. When rotating the camshaft with the timing chain removed, rotate it after lowering the piston from the top dead center position.**
- **When rotating the camshaft using a wrench on the cast hexagon, the wrench may contact the rocker arm and damage the rocker arm. To prevent damage to the rocker arm when holding the camshaft on the cast hexagon, use a wrench on the rear side of the engine as shown in the figure to secure a clearance between the cam.**
- **When installing the oil pan, refer to the Oil Pan Installation Note and apply the silicone sealant correctly. If the silicone sealant application position is incorrect or there is any break in the application track, engine oil leakage will occur due to a poor seal.**

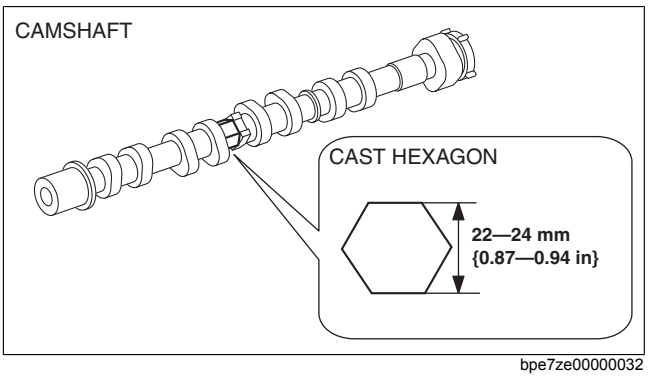



am3uuw00008968

**Note**

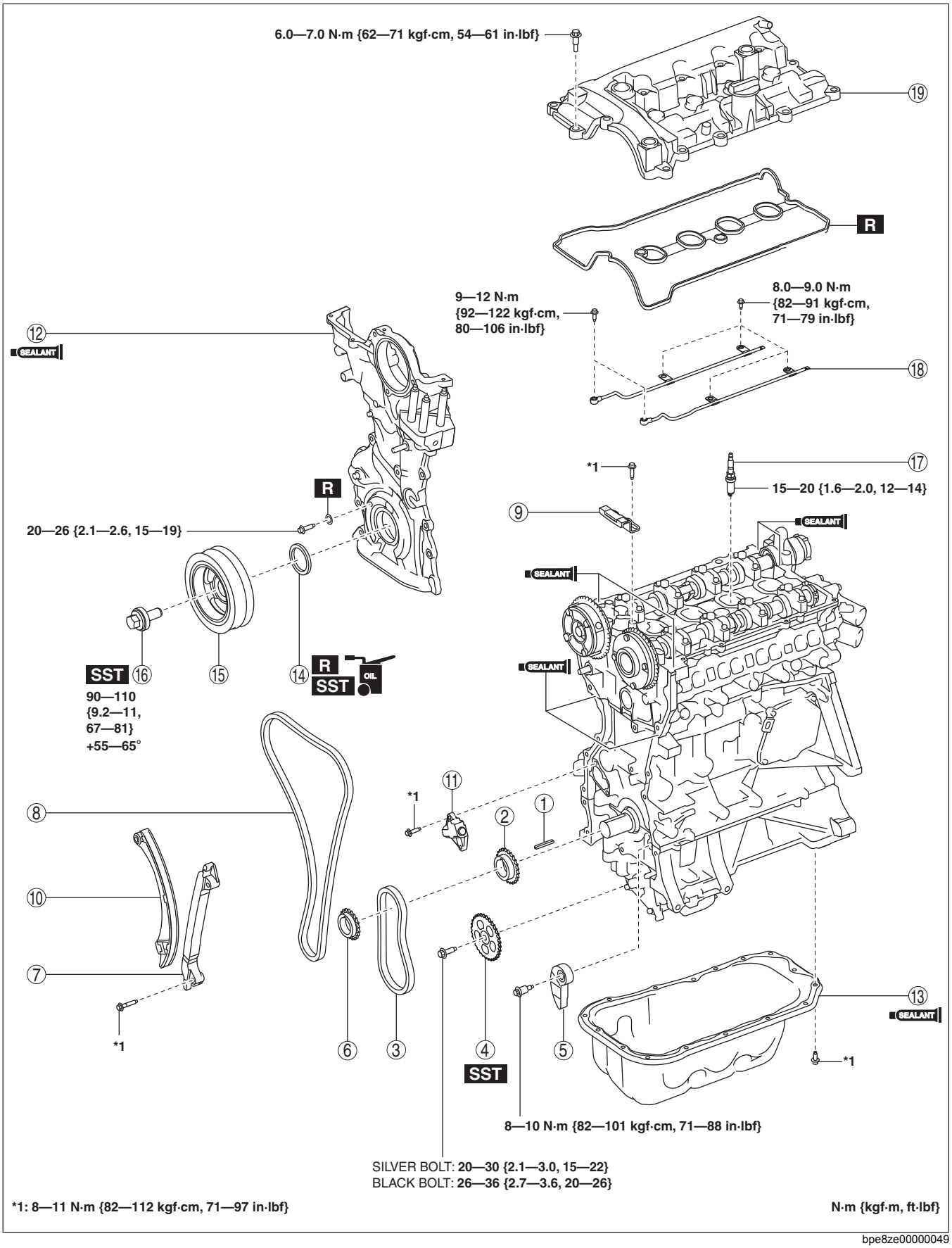
- Width at the cast hexagon of the camshaft is **22—24 mm {0.87—0.94 in}**.

1. Assemble in the order indicated in the table.




bpe7ze00000032

## SKYACTIV-G 1.3, SKYACTIV-G 1.5 (Except MX-5 (ND)), SKYACTIV-G 2.0 (Except MX-5 (ND))



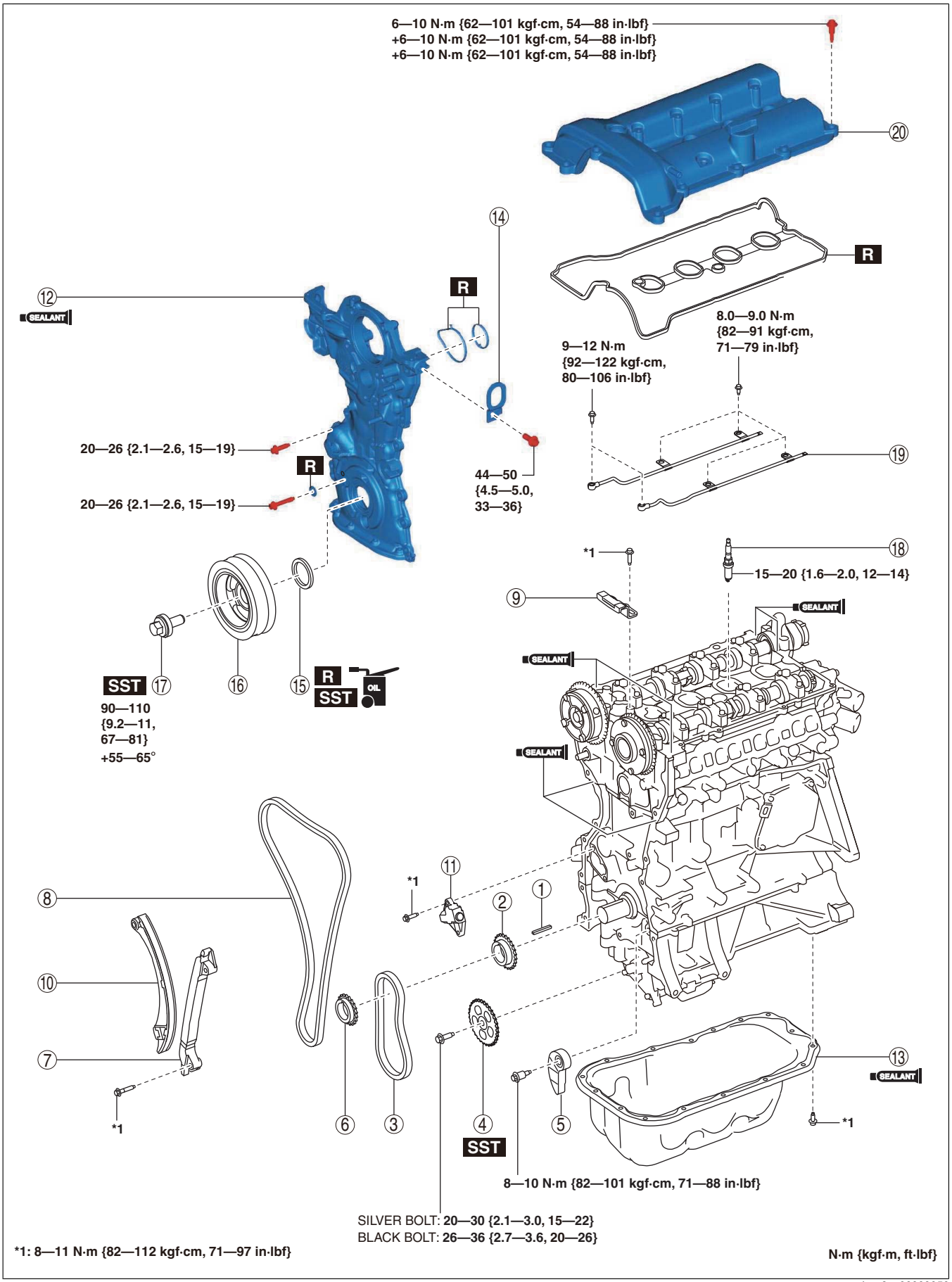

bpe8ze00000049

| | |
|---|---|
| 1 | Key |
| 2 | Oil pump drive sprocket |
| 3 | Oil pump chain |

| | |
|---|---|
| 4 | Oil pump driven sprocket<br>(See Oil Pump Driven Sprocket Assembly Note (SKYACTIV-G 1.3, SKYACTIV-G 1.5, SKYACTIV-G 2.0).) |
| 5 | Oil pump chain tensioner |
| 6 | Crankshaft sprocket |
| 7 | Chain guide (No.2) |
| 8 | Timing chain<br>(See Timing Chain Assembly Note.) |
| 9 | Chain guide (No.1)<br>(See Timing Chain Assembly Note.) |
| 10 | Tensioner arm<br>(See Timing Chain Assembly Note.) |
| 11 | Chain tensioner<br>(See Timing Chain Assembly Note.) |
| 12 | Engine front cover<br>(See Engine Front Cover Assembly Note (Except MX-5 (ND)).) |

| | |
|---|---|
| 13 | Oil pan<br>(See Oil Pan Assembly Note (Except MX-5 (ND)).) |
| 14 | Front oil seal<br>(See Front Oil Seal Assembly Note.) |
| 15 | Crankshaft pulley |
| 16 | Crankshaft pulley lock bolt<br>(See Crankshaft Pulley Lock Bolt Assembly Note.) |
| 17 | Spark plug |
| 18 | Oil shower pipe<br>(See Oil Shower Pipe Assembly Note.) |
| 19 | Cylinder head cover<br>(See Cylinder Head Cover Assembly Note (Except MX-5 (ND)).) |

## SKYACTIV-G 1.5 (MX-5 (ND)), SKYACTIV-G 2.0 (MX-5 (ND))



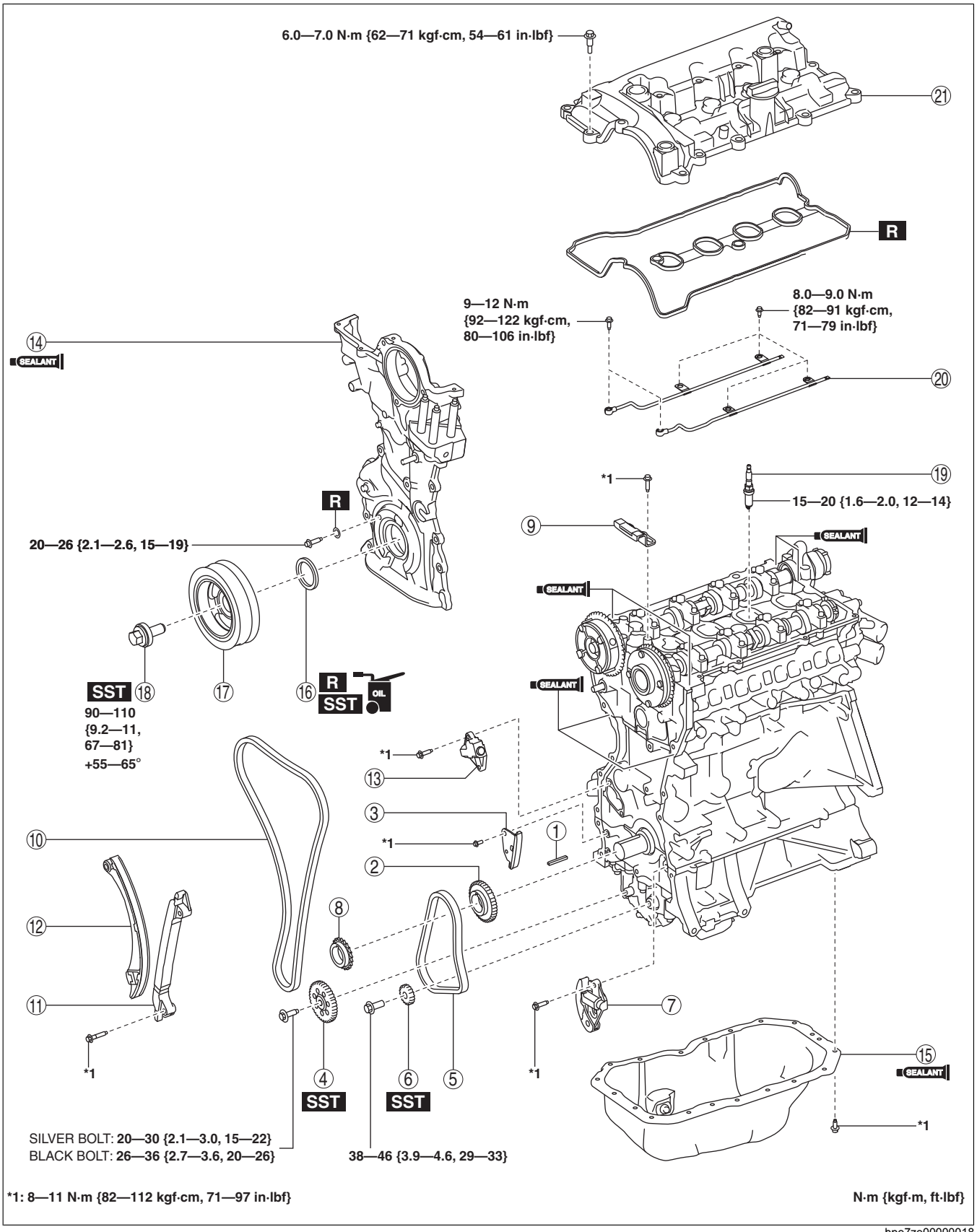

bpe8ze00000050

| | |
|---|---|
| 1 | Key |
| 2 | Oil pump drive sprocket |

| | |
|---|---|
| 3 | Oil pump chain |
| 4 | Oil pump driven sprocket<br>(See Oil Pump Driven Sprocket Assembly Note (SKYACTIV-G 1.3, SKYACTIV-G 1.5, SKYACTIV-G 2.0).) |
| 5 | Oil pump chain tensioner |
| 6 | Crankshaft sprocket |
| 7 | Chain guide (No.2)<br>(See Timing Chain Assembly Note.) |
| 8 | Timing chain<br>(See Timing Chain Assembly Note.) |
| 9 | Chain guide (No.1)<br>(See Timing Chain Assembly Note.) |
| 10 | Tensioner arm<br>(See Timing Chain Assembly Note.) |
| 11 | Chain tensioner<br>(See Timing Chain Assembly Note.) |
| 12 | Engine front cover<br>(See Engine Front Cover Assembly Note (MX-5 (ND)).) |

| | |
|---|---|
| 13 | Oil pan<br>(See Oil Pan Assembly Note (MX-5 (ND)).) |
| 14 | Engine hanger |
| 15 | Front oil seal<br>(See Front Oil Seal Assembly Note.) |
| 16 | Crankshaft pulley |
| 17 | Crankshaft pulley lock bolt<br>(See Crankshaft Pulley Lock Bolt Assembly Note.) |
| 18 | Spark plug |
| 19 | Oil shower pipe<br>(See Oil Shower Pipe Assembly Note.) |
| 20 | Cylinder head cover<br>(See Cylinder Head Cover Installation Note (MX-5 (ND)).) |

## SKYACTIV-G 2.5




bpe7ze00000018

| | |
|---|---|
| 1 | Key |
| 2 | Oil pump drive sprocket |
| 3 | Oil pump chain guide |

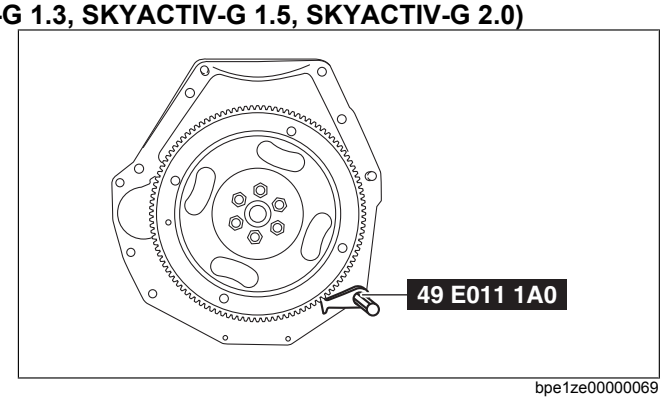
| | |
|---|---|
| 4 | Oil pump driven sprocket<br>(See Oil Pump Chain Assembly Note (SKYACTIV-G 2.5).) |

| | |
|---|---|
| 5 | Oil pump chain<br>(See Oil Pump Chain Assembly Note (SKYACTIV-G 2.5).) |
| 6 | Balancer shaft sprocket<br>(See Oil Pump Chain Assembly Note (SKYACTIV-G 2.5).) |
| 7 | Oil pump chain tensioner<br>(See Oil Pump Chain Assembly Note (SKYACTIV-G 2.5).) |
| 8 | Crankshaft sprocket |
| 9 | Chain guide (No.2) |
| 10 | Timing chain<br>(See Timing Chain Assembly Note.) |
| 11 | Chain guide (No.1)<br>(See Timing Chain Assembly Note.) |
| 12 | Tensioner arm<br>(See Timing Chain Assembly Note.) |
| 13 | Chain tensioner<br>(See Timing Chain Assembly Note.) |
| 14 | Engine front cover<br>(See Engine Front Cover Assembly Note (Except MX-5 (ND)).) |
| 15 | Oil pan<br>(See Oil Pan Assembly Note (Except MX-5 (ND)).) |
| 16 | Front oil seal<br>(See Front Oil Seal Assembly Note.) |
| 17 | Crankshaft pulley |
| 18 | Crankshaft pulley lock bolt<br>(See Crankshaft Pulley Lock Bolt Assembly Note.) |
| 19 | Spark plug |
| 20 | Oil shower pipe<br>(See Oil Shower Pipe Assembly Note.) |
| 21 | Cylinder head cover<br>(See Cylinder Head Cover Assembly Note (Except MX-5 (ND)).) |

**Oil Pump Driven Sprocket Assembly Note (SKYACTIV-G 1.3, SKYACTIV-G 1.5, SKYACTIV-G 2.0)**

1. Hold the crankshaft using the **SST**.
2. Install the oil pump driven sprocket.

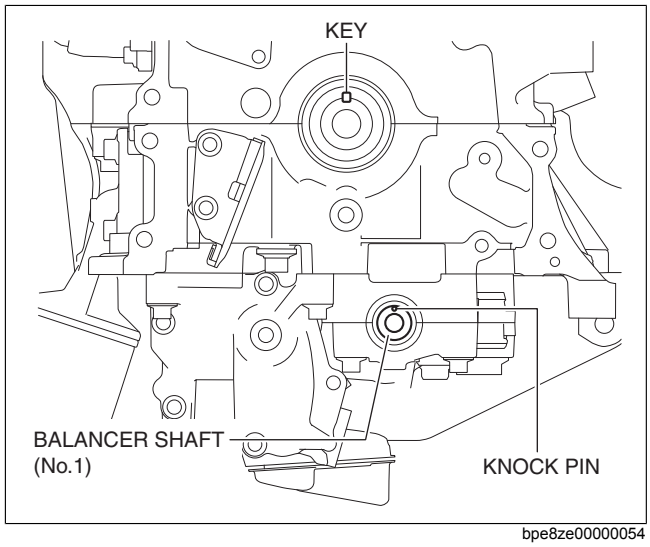
**Tightening torque**
**Silver bolt: 20—30 N·m {2.1—3.0 kgf·m, 15—22 ft·lbf}**
**Black bolt: 26—36 N·m {2.7—3.6 kgf·m, 20—26 ft·lbf}**



bpe1ze00000069

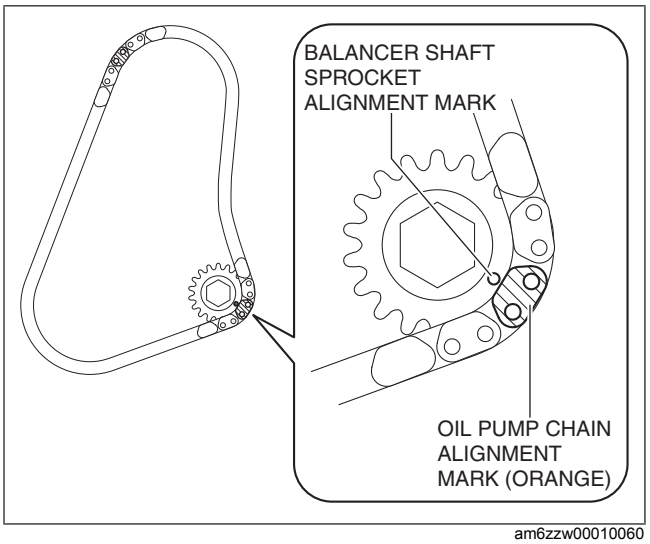
**Oil Pump Chain Assembly Note (SKYACTIV-G 2.5)**

1. Verify that the key and knock pin are aligned to the positions shown in the figure.
2. Temporarily assemble the oil pump driven sprocket.
3. Temporarily tighten the oil pump driven sprocket installation bolt.



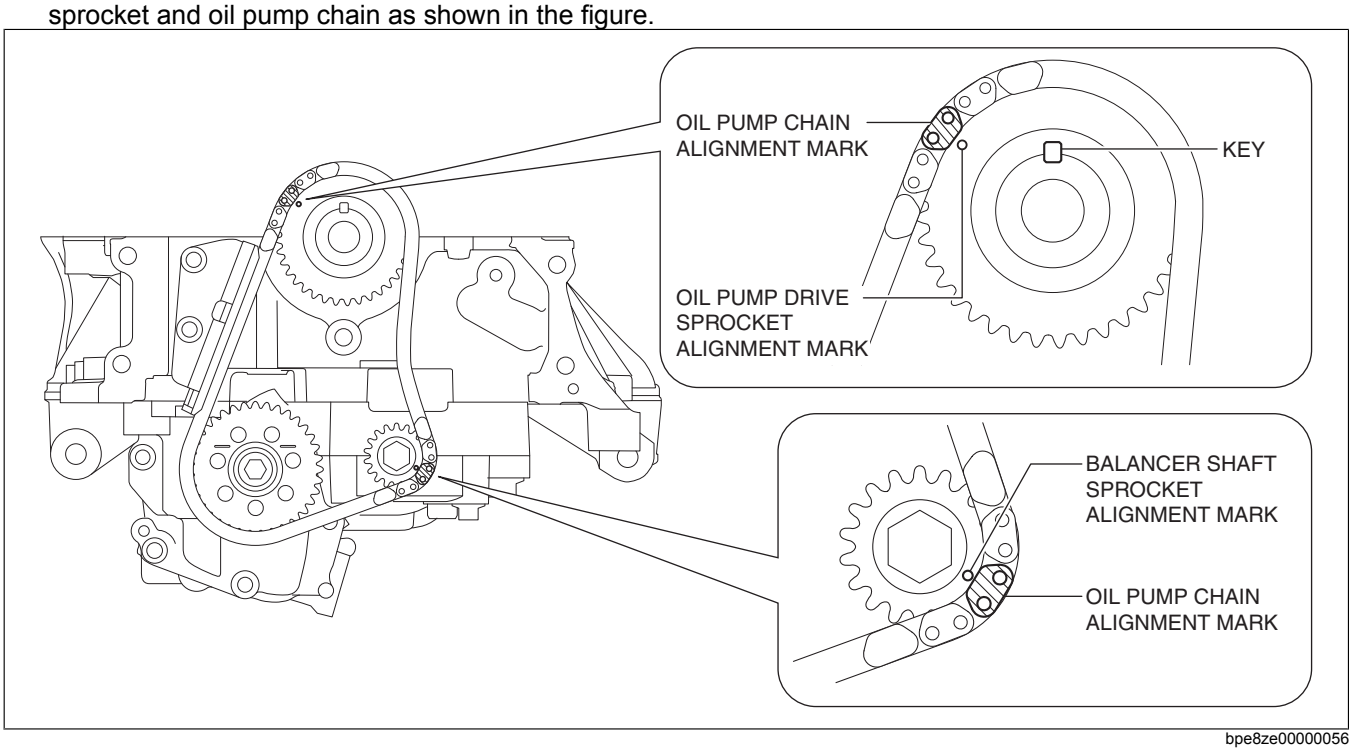
bpe8ze00000054

4. Align the oil pump chain alignment mark with the balancer shaft sprocket alignment mark.




am6zzw00010060

5. Install the oil pump chain and balancer shaft sprocket as a single unit while aligning the alignment marks on each sprocket and oil pump chain as shown in the figure.




bpe8ze00000056

6. Temporarily tighten the balancer shaft sprocket installation bolt.
7. Install the oil pump chain tensioner.

**Tightening torque**
**8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}**

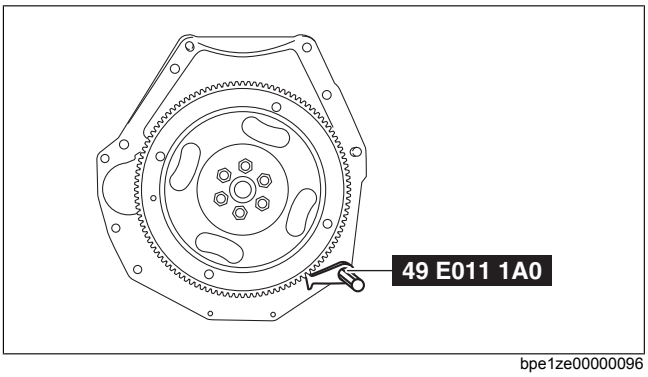
**Caution**
- **At this stage, do not remove the wire or paper clip installed to the oil pump chain tensioner.**

8. Hold the crankshaft using the **SST**.
9. Tighten the oil pump driven sprocket installation bolt.

   **Tightening torque**
   **Silver bolt: 20—30 N·m {2.1—3.0 kgf·m, 15—22 ft·lbf}**
   **Black bolt: 26—36 N·m {2.7—3.6 kgf·m, 20—26 ft·lbf}**

10. Tighten the balancer shaft sprocket installation bolt.

    **Tightening torque**
    **38—46 N·m {3.9—4.6 kgf·m, 29—33 ft·lbf}**

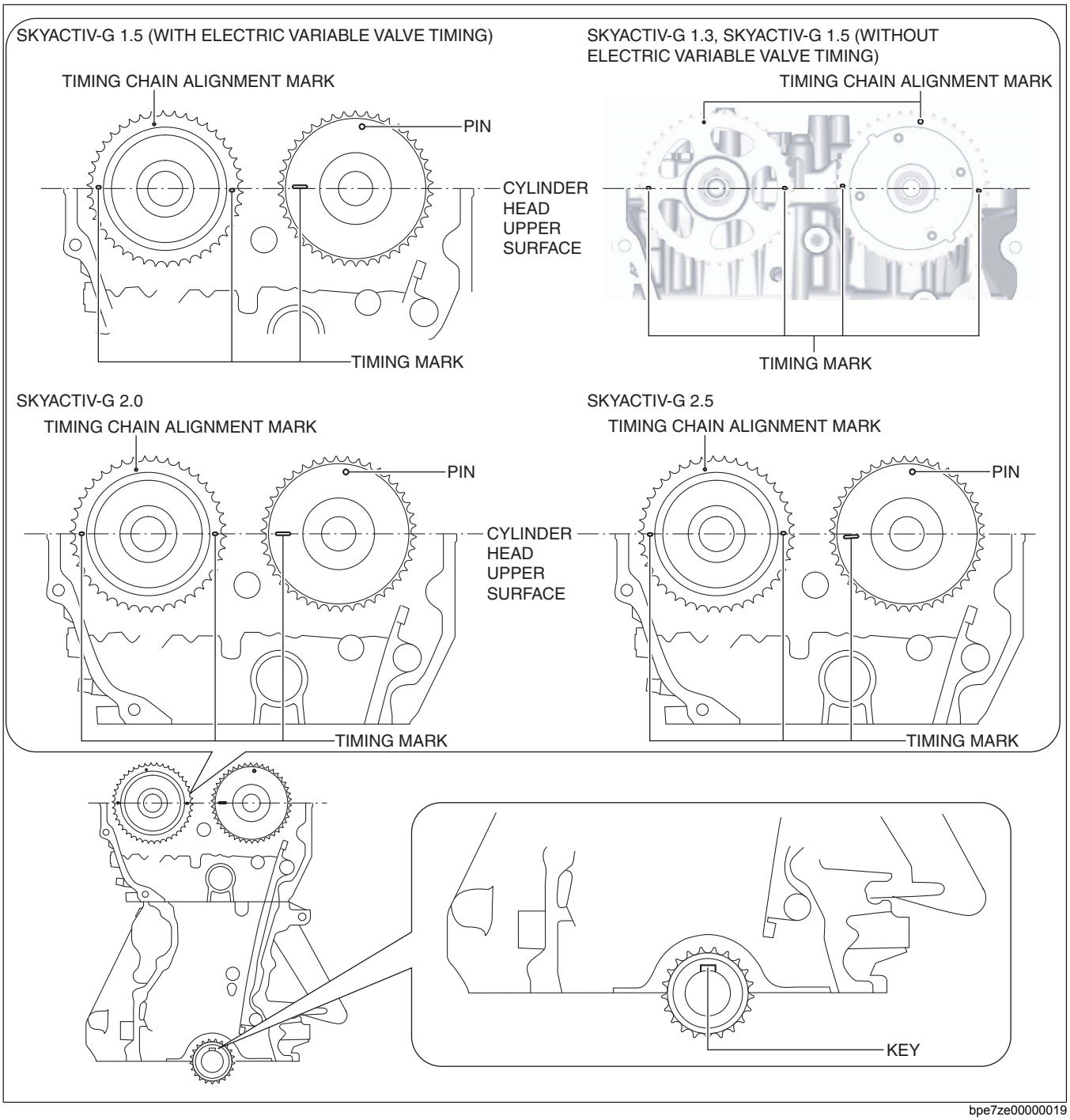


bpe1ze00000096

11. Remove the wire or paper clip installed to the oil pump chain tensioner and apply tension to the oil pump chain.
    - If a new oil pump chain tensioner is used, remove the installed stopper.

**Timing Chain Assembly Note**

1. Verify that the timing marks and the key are aligned to the position shown in the figure.
   - If they are not in the position shown in the figure, rotate the camshaft and crankshaft to set the cylinder No. 1 top dead center (TDC).



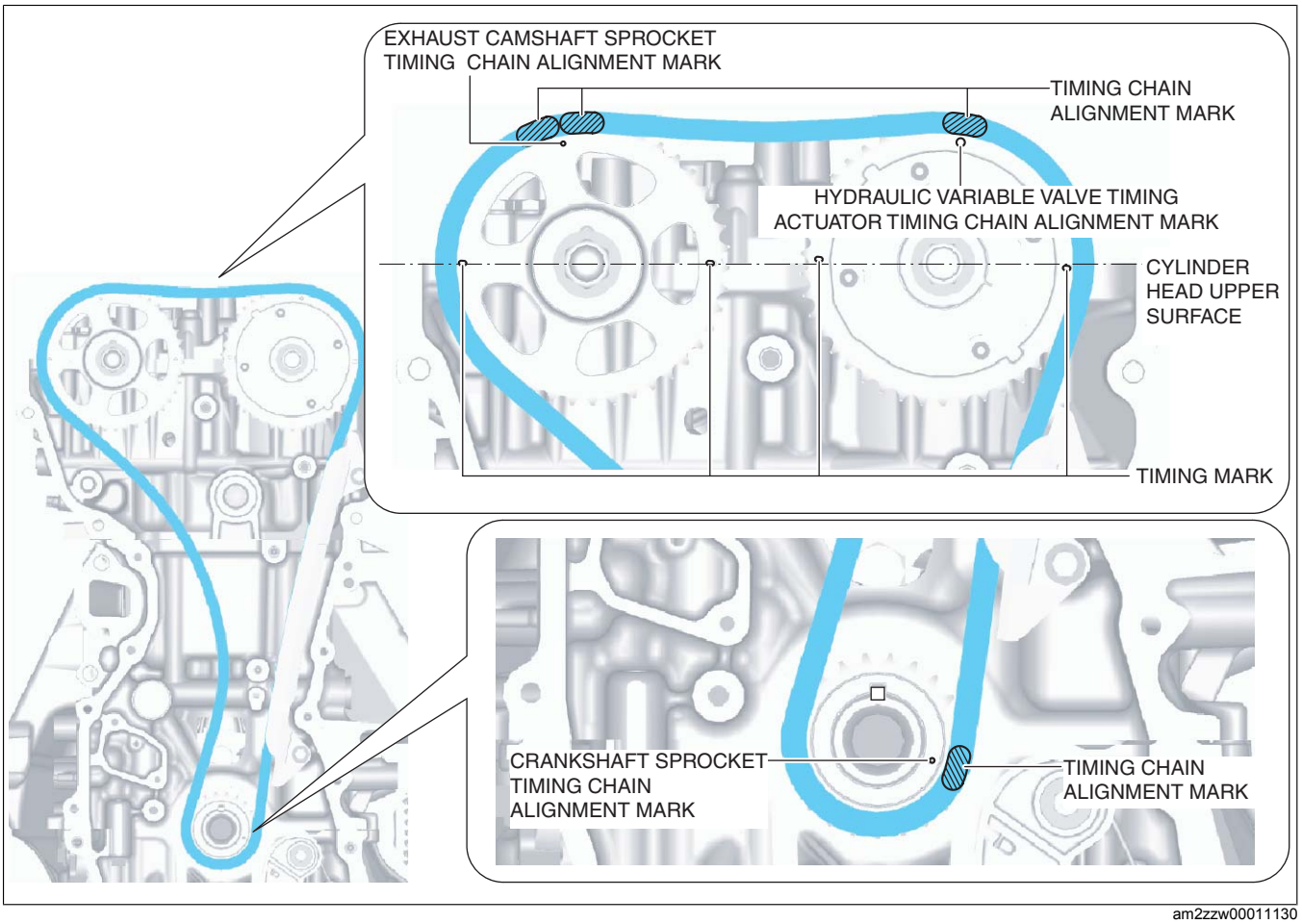

bpe7ze00000019

**Note**

- The timing mark of SKYACTIV-G 1.3, SKYACTIV-G 1.5 and SKYACTIV-G 2.5 is not parallel with the upper surface of the cylinder head.

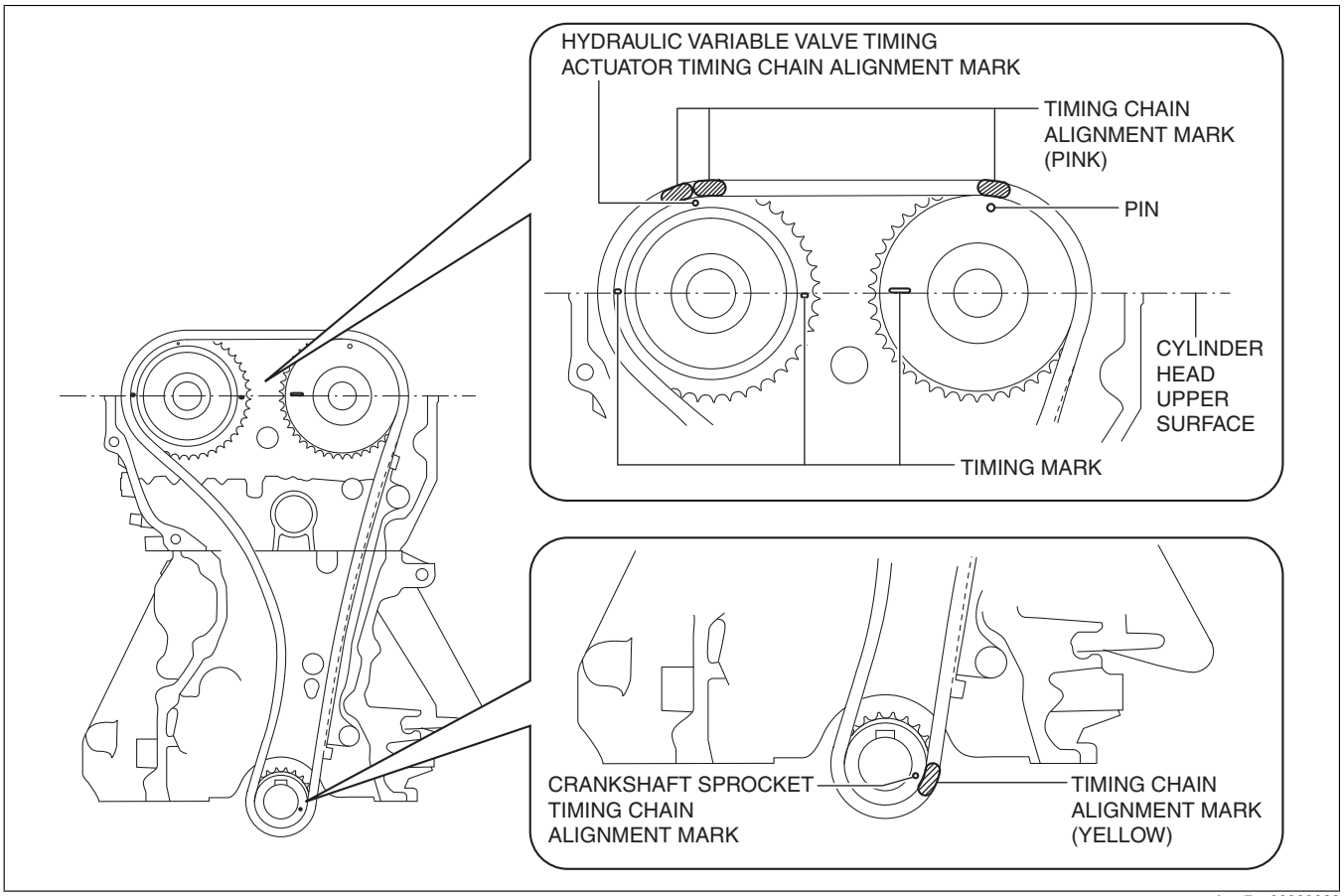
2. Install the timing chain while aligning the marks on each sprocket and the timing chain as shown in the figure.

## SKYACTIV-G 1.3, SKYACTIV-G 1.5 (without electric variable valve timing)



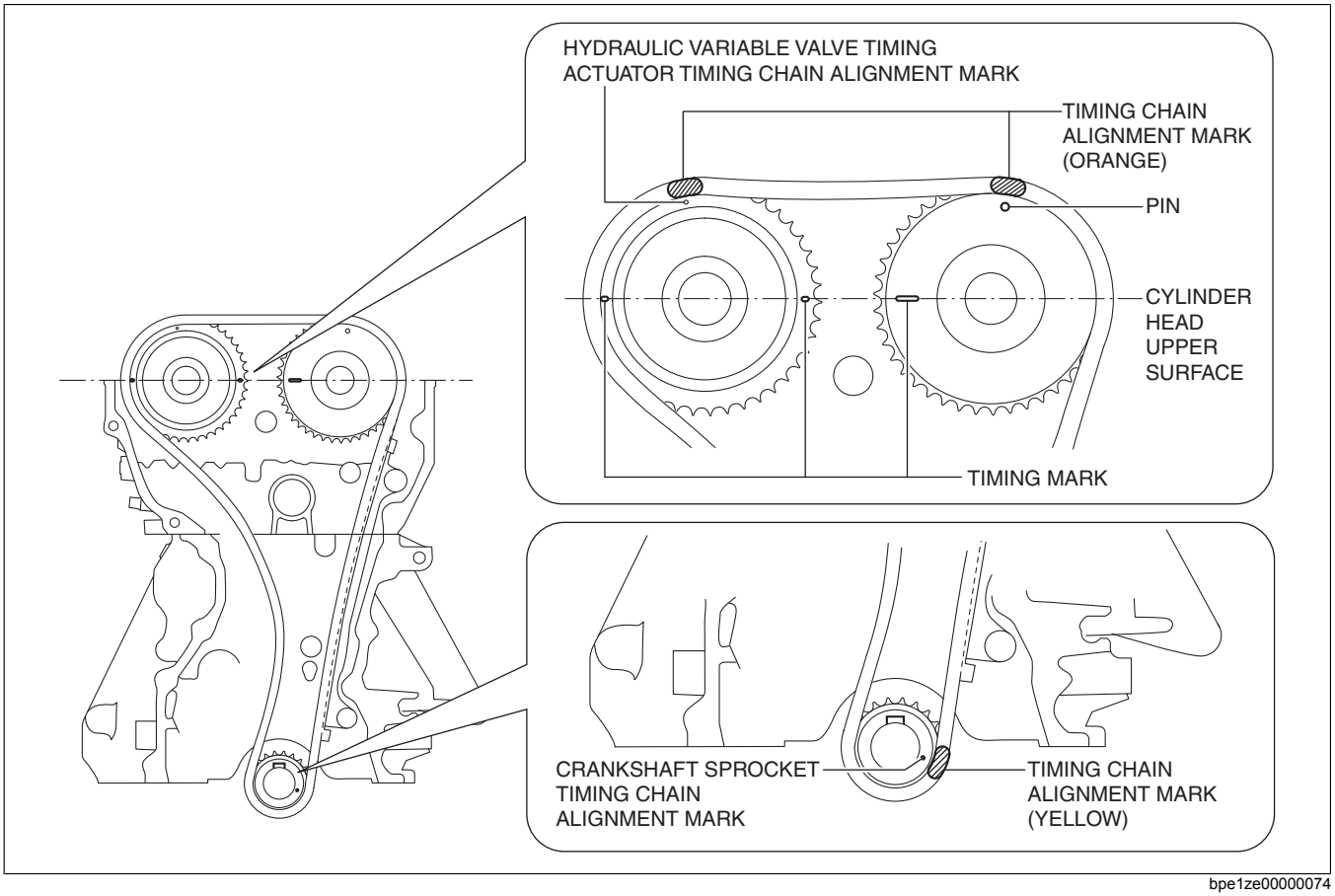

am2zzw00011130

**SKYACTIV-G 1.5 (with electric variable valve timing)**



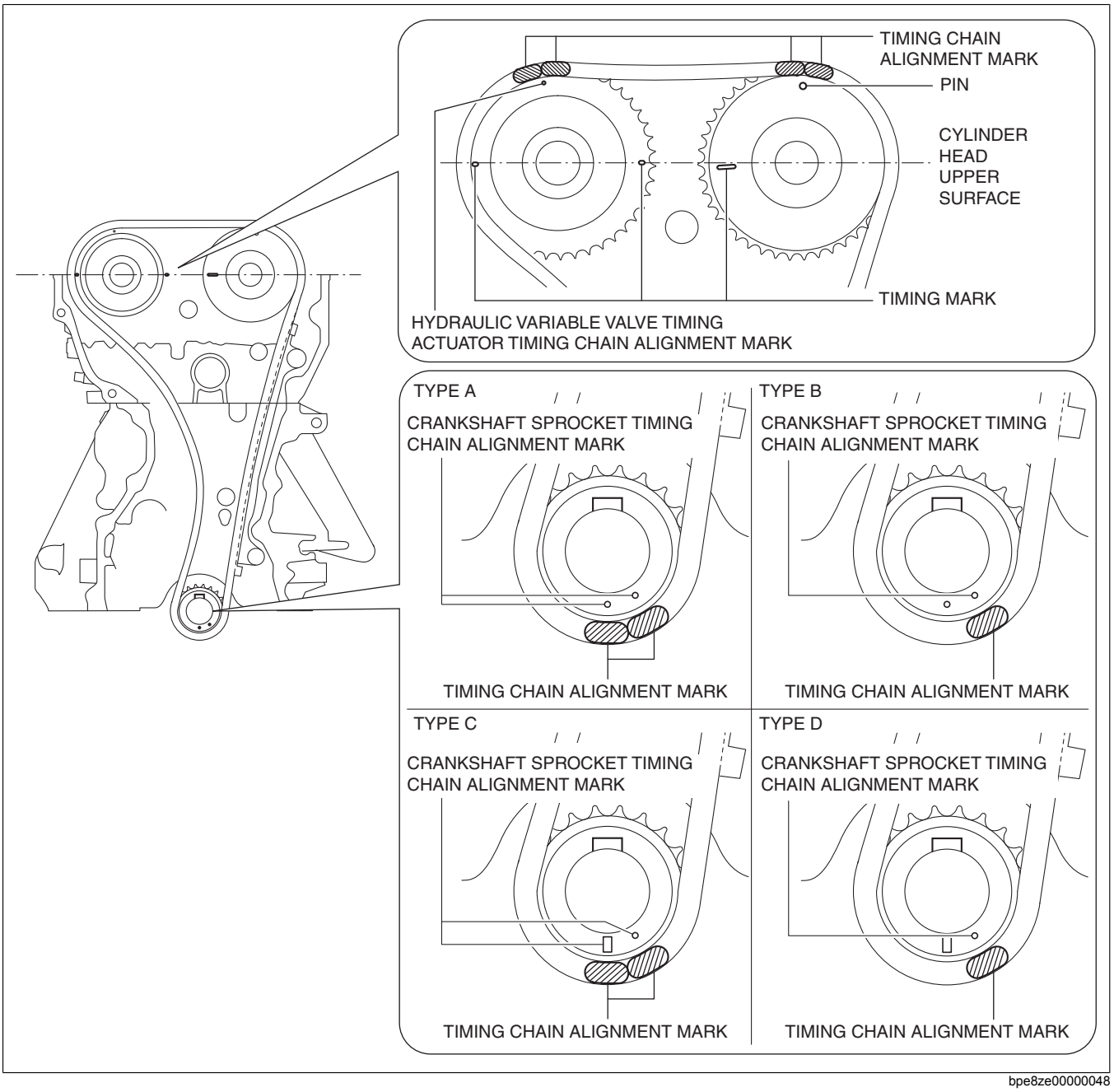

bpe7ze00000020

**SKYACTIV-G 2.0**




bpe1ze00000074

**SKYACTIV-G 2.5**




bpe8ze00000048

3. Install the chain guide (No.1).

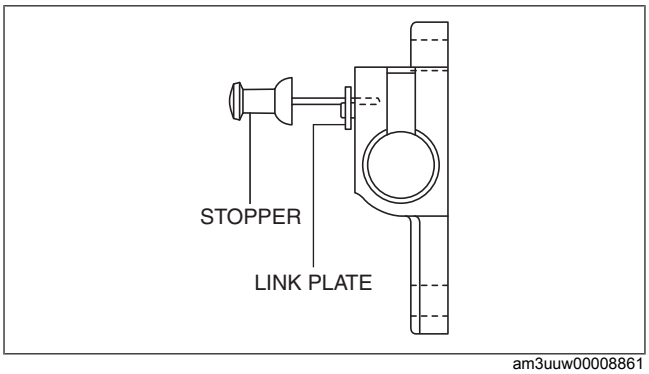
   **Tightening torque**
   **8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}**

4. Install the tensioner arm.
5. Install the chain tensioner.

   **Tightening torque**
   **8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}**

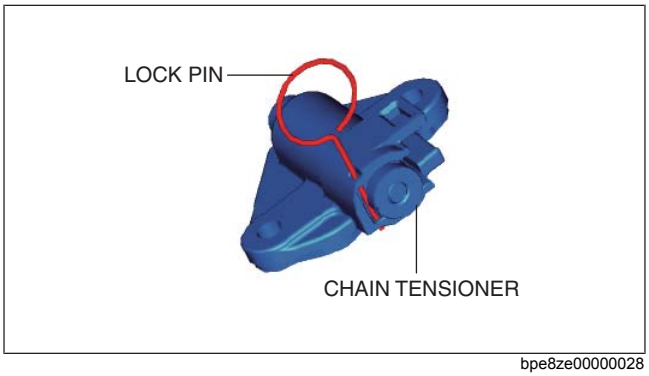
6. After installing the chain tensioner, remove the installed wire or paper clip, and then apply tension to the timing chain. (chain tensioner (type A))

- If a new chain tensioner is used, remove the installed stopper.



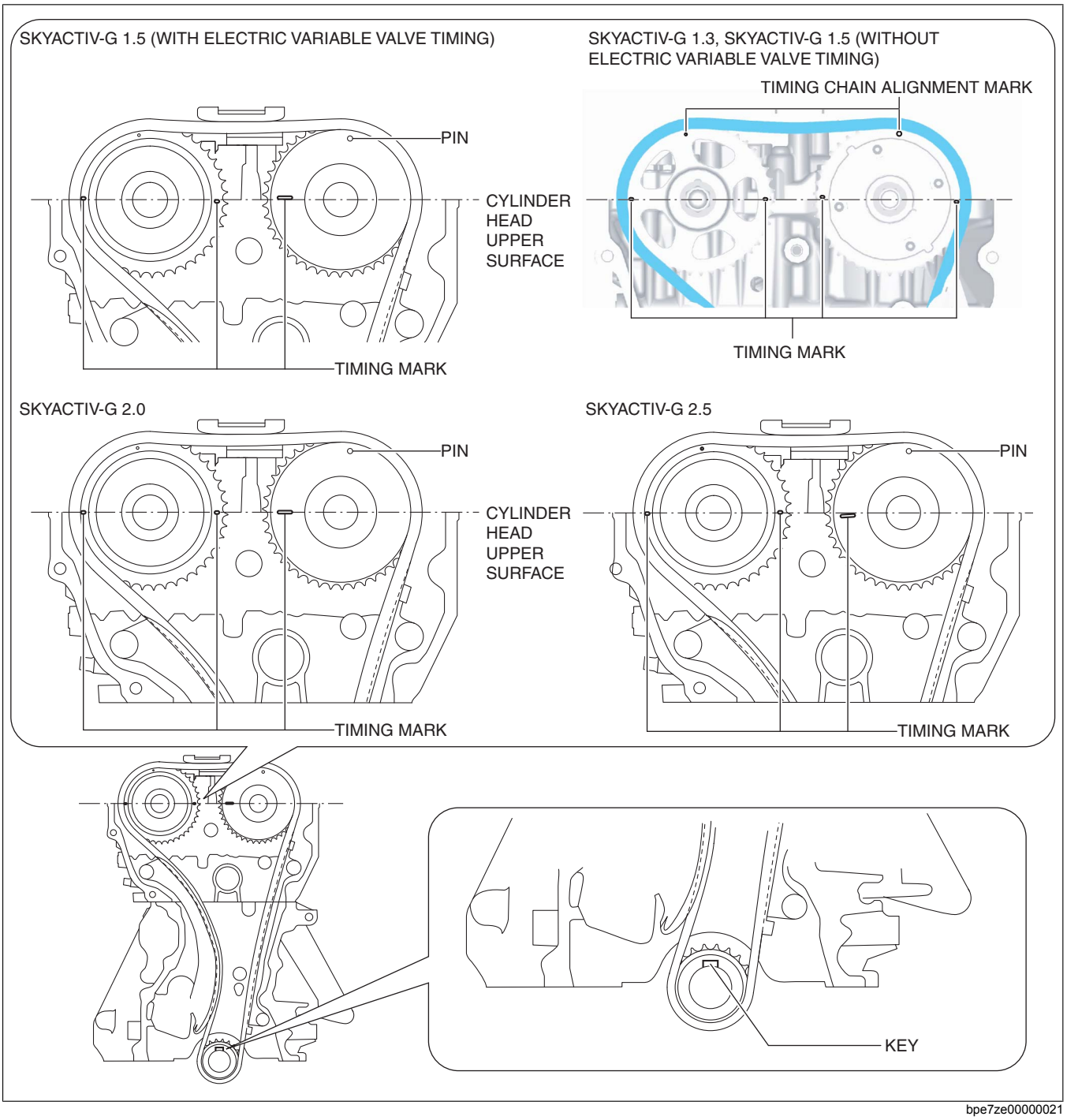
am3uuw00008861

7. After installing the timing chain tensioner, remove the installed rod, and then apply tension to the timing chain. (chain tensioner (type B))
   - If a new chain tensioner is used, remove the installed lock pin.
8. Verify that there is no looseness in the timing chain, and re-verify that each sprocket is in the specified location.



bpe8ze00000028

9. Rotate the crankshaft clockwise two turns and inspect the valve timing.



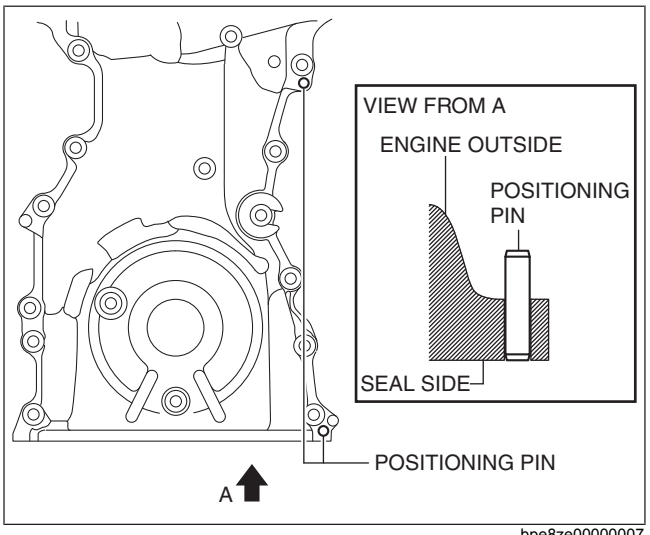

bpe7ze00000021

**Note**

- The timing mark of SKYACTIV-G 1.3, SKYACTIV-G 1.5 and SKYACTIV-G 2.5 is not parallel with the upper surface of the cylinder head.

**Engine Front Cover Assembly Note (Except MX-5 (ND))**

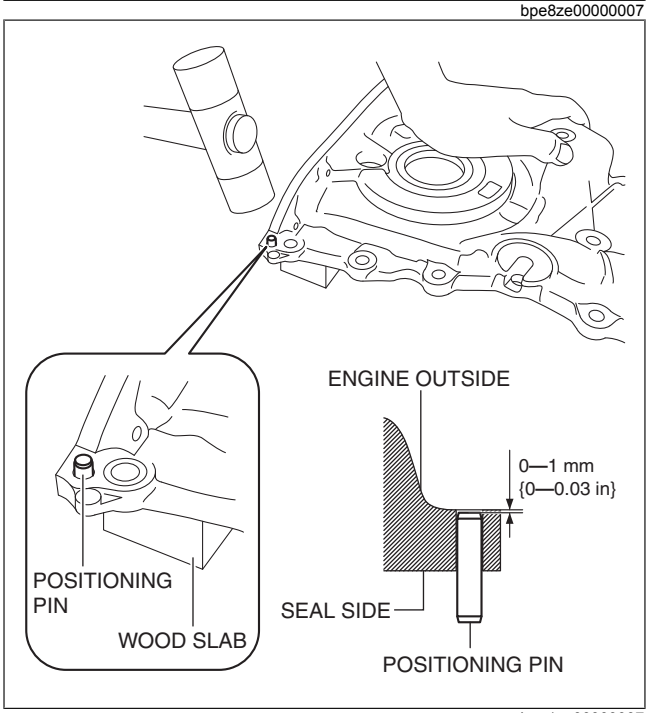
**Note**
- For a new engine front cover, the positioning pins in the two locations shown in the figure project to the outside of the engine.



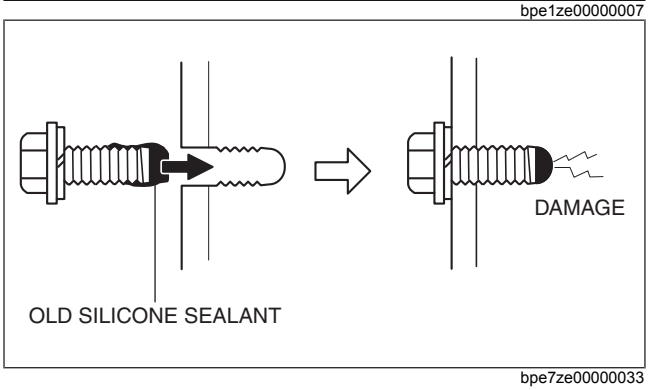
bpe8ze00000007

1. If the engine front cover is newly replaced, tap the positioning pins in the two locations to the seal surface side.

**Caution**
- **If a bolt with silicone sealant adhering to it is used, it could result in cracks in the cylinder head and cylinder block.**



bpe1ze00000007



bpe7ze00000033

2. When reusing the engine front cover installation bolts, remove sealant adhering to the bolts.

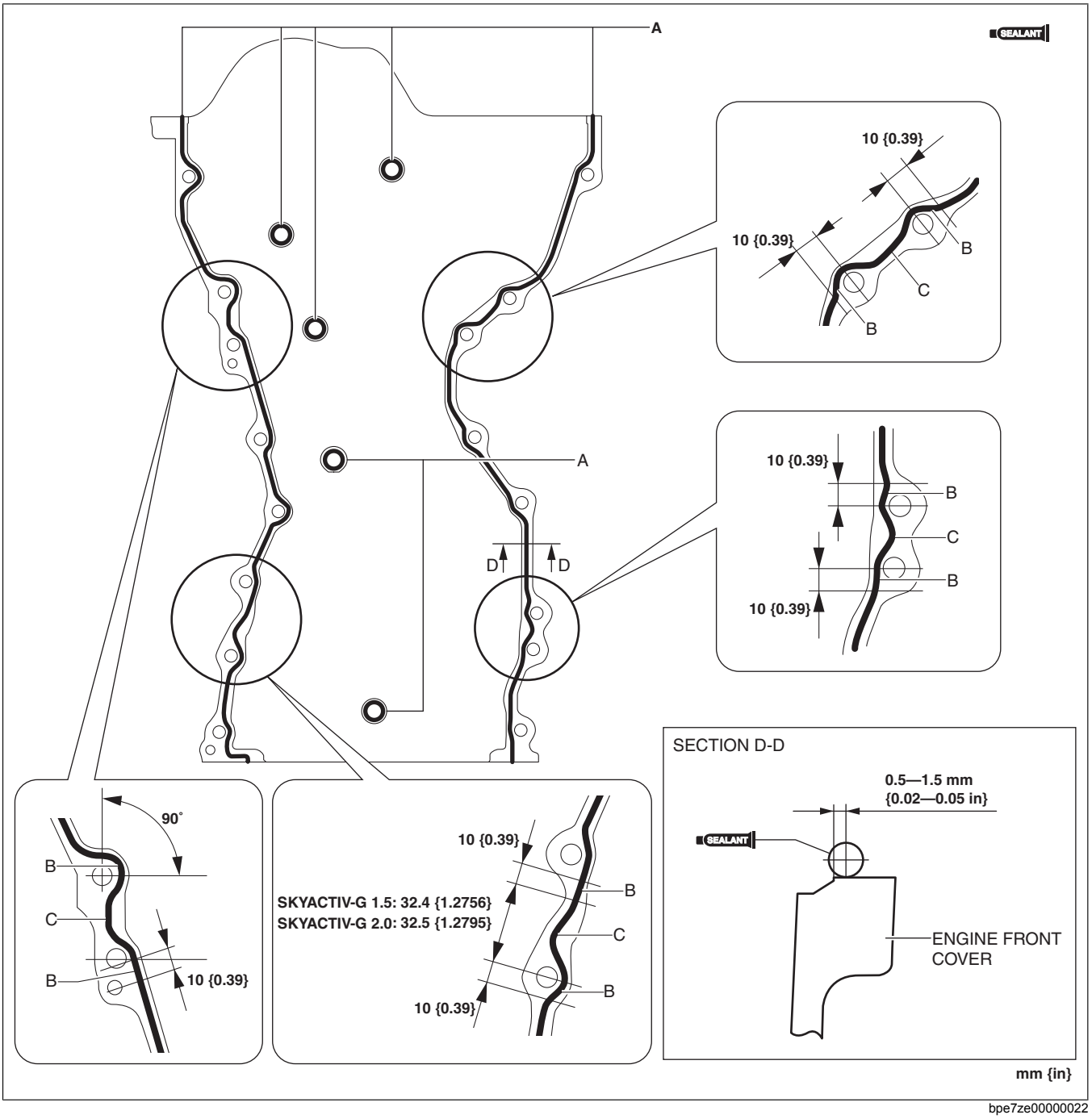
**Caution**
- **If oil, dirt and silicone sealant remains on the silicone sealant application area, the silicone sealant will not seal which will cause oil leakage.**

3. Completely clean and remove oil, dirt, silicone sealant or other foreign matter that may be adhering to the engine front cover, cylinder head, and cylinder block.

**Caution**
- **Apply the silicone sealant in a single, unbroken line.**
- **To prevent silicone sealant from hardening, adhere the engine front cover and the cylinder block firmly within 10 min. after applying silicone sealant. After adhering them, tighten the installation bolts immediately.**

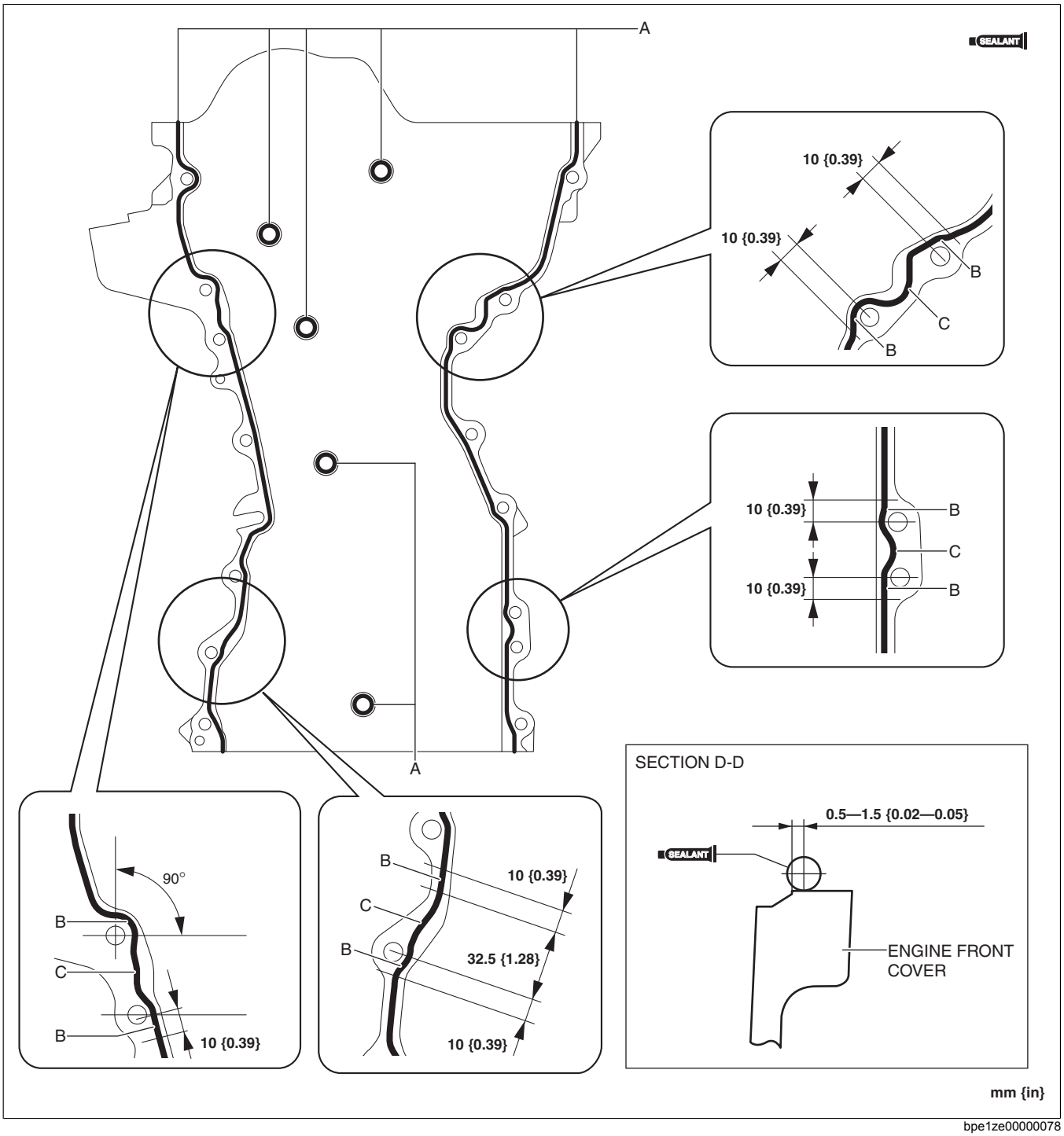
4. Apply silicone sealant (TB1217D or equivalent) to the engine front cover as shown in the figure.

**SKYACTIV-G 1.3, SKYACTIV-G 1.5, SKYACTIV-G 2.0**




bpe7ze00000022

**SKYACTIV-G 2.5**



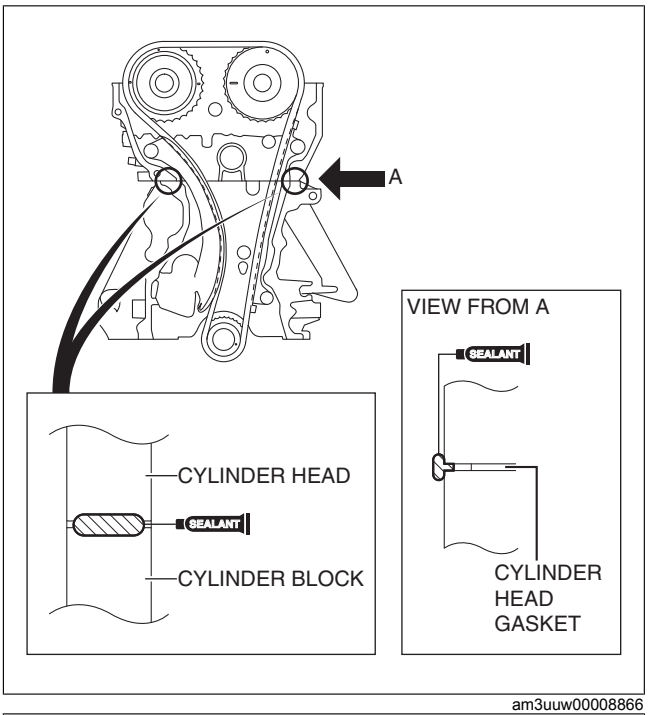

bpe1ze00000078

**Thickness**
**A: 2—6 mm {0.1—0.2 in}**
**B: 4—6 mm {0.16—0.23 in}**
**C: 4—8 mm {0.2—0.3 in}**

**Caution**
- **Apply the silicone sealant so that it goes into the cylinder head gasket.**

5. Apply silicone sealant (TB1217D or equivalent) to the areas shown in the figure.
6. Install the engine front cover to the engine.

**Note**
- Temporarily install an appropriate bolt to the drive belt auto tensioner installation bolt hole to prevent:
  — A silicone sealant adhesion malfunction in the drive belt auto tensioner installation bolt hole.
  — A bolt mis-installation due to silicone sealant hardening.

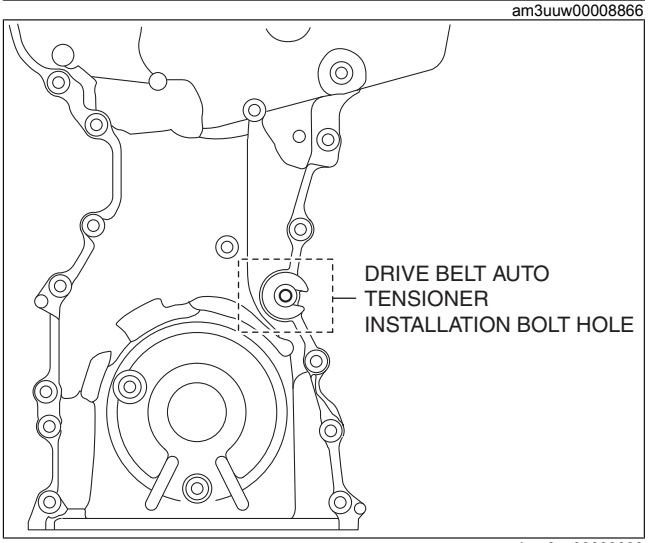


am3uuw00008866

7. Prepare an appropriate **M8 X 1.25** bolt (length **40 mm {1.6 in}**).

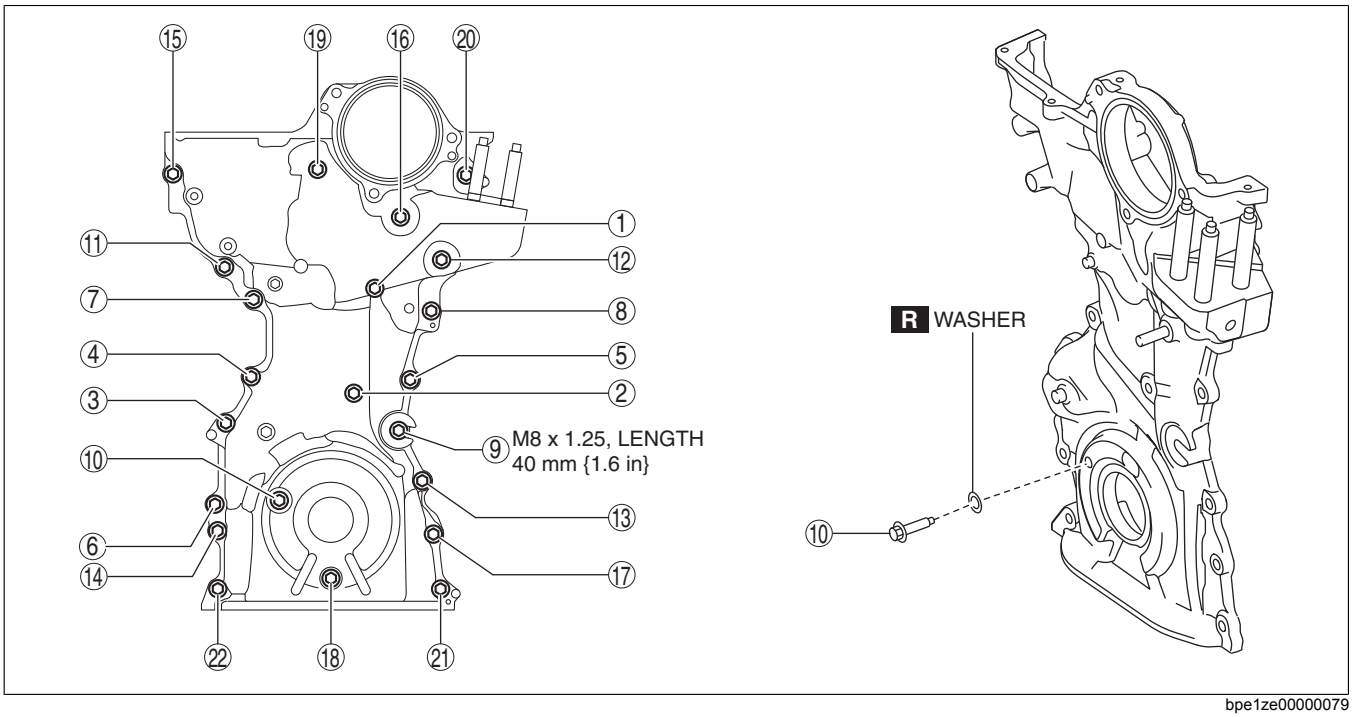
**Caution**
- **For the number 10 bolt of the tightening order, install the bolt with new washer.**

8. Tighten the engine front cover installation bolts in the order shown in the figure.



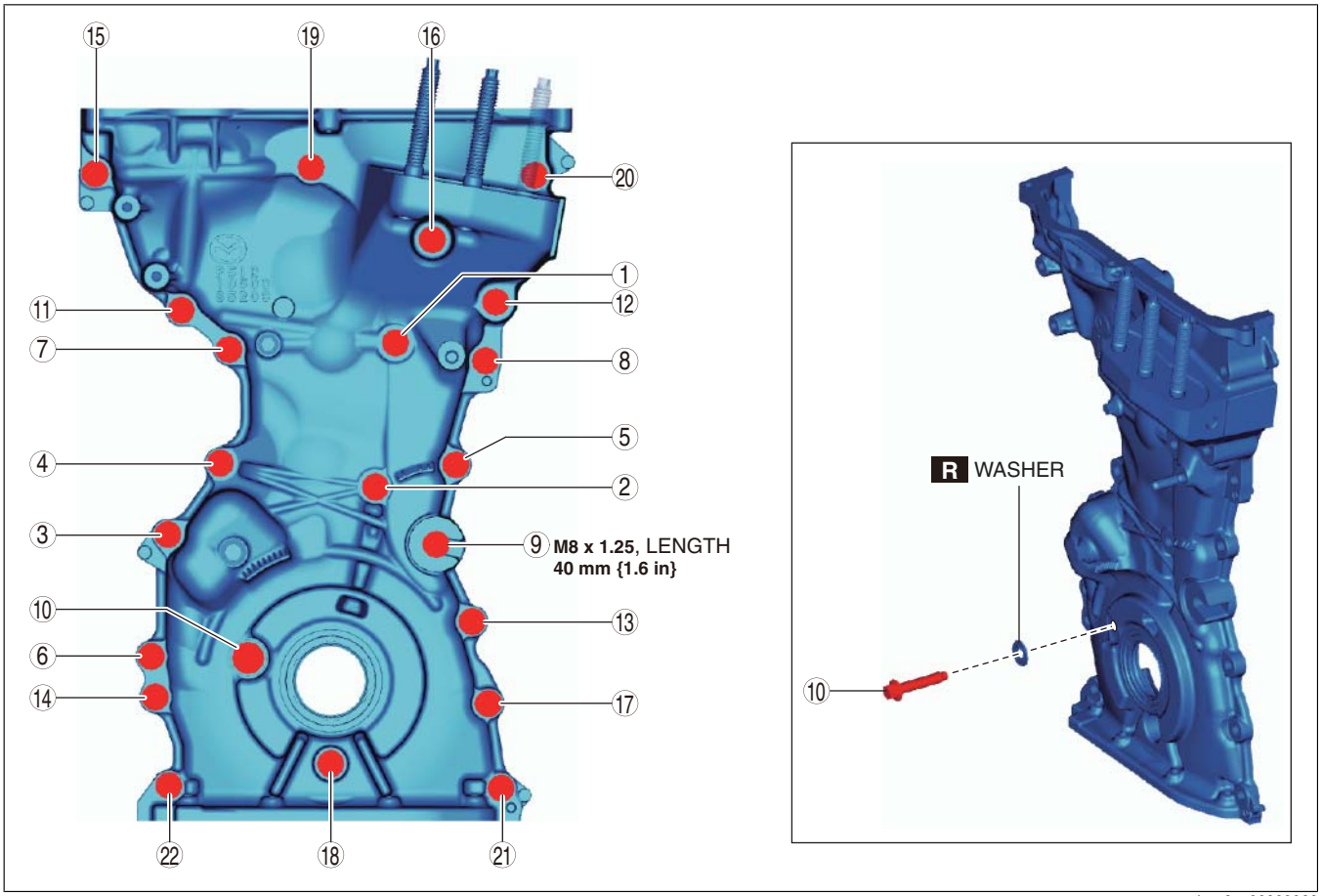
bpe8ze00000008

**With electric variable valve timing**




bpe1ze00000079

**Without electric variable valve timing**



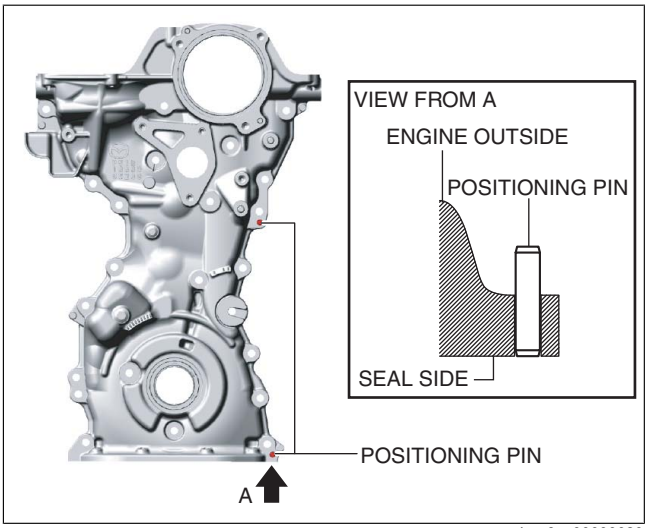

bpe8ze00000060

**Tightening torque**
**20—26 N·m {2.1—2.6 kgf·cm, 15—19 in·lbf}**

9. Remove the bolt installed to the drive belt auto tensioner installation bolt hole when installing the drive belt auto tensioner.

**Engine Front Cover Assembly Note (MX-5 (ND))**

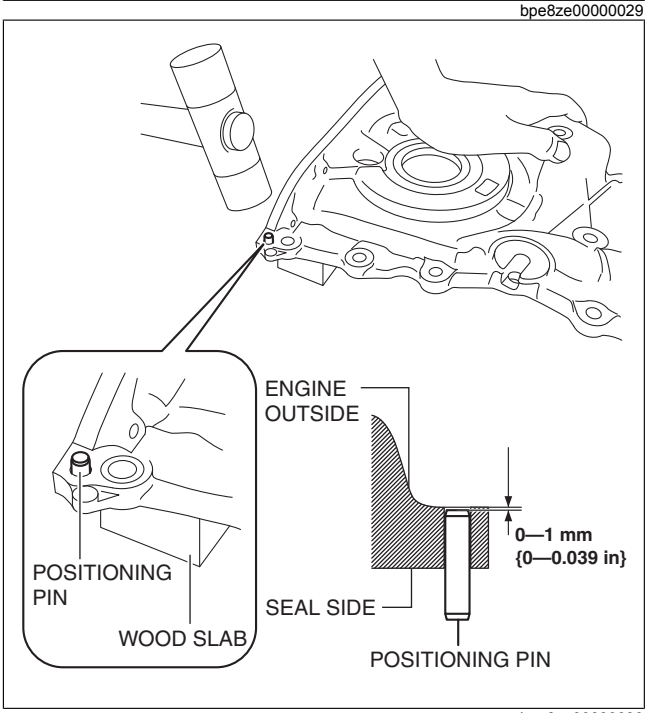
**Note**
- For a new engine front cover, the positioning pins in the two locations shown in the figure project to the outside of the engine.



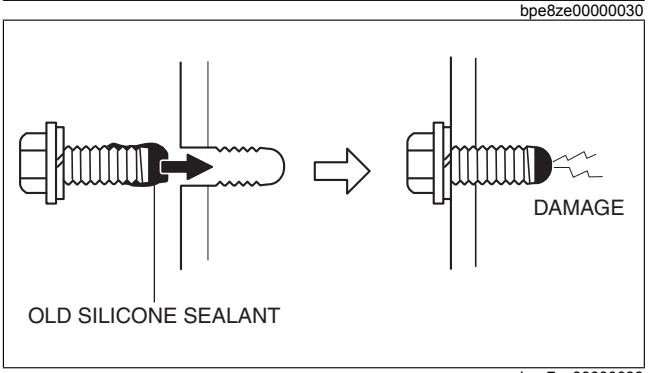
bpe8ze00000029

1. If the engine front cover is newly replaced, tap the positioning pins in the **two locations** to the seal surface side.

**Caution**
- **If a bolt with silicone sealant adhering to it is used, it could result in cracks in the cylinder head and cylinder block.**



bpe8ze00000030



bpe7ze00000033

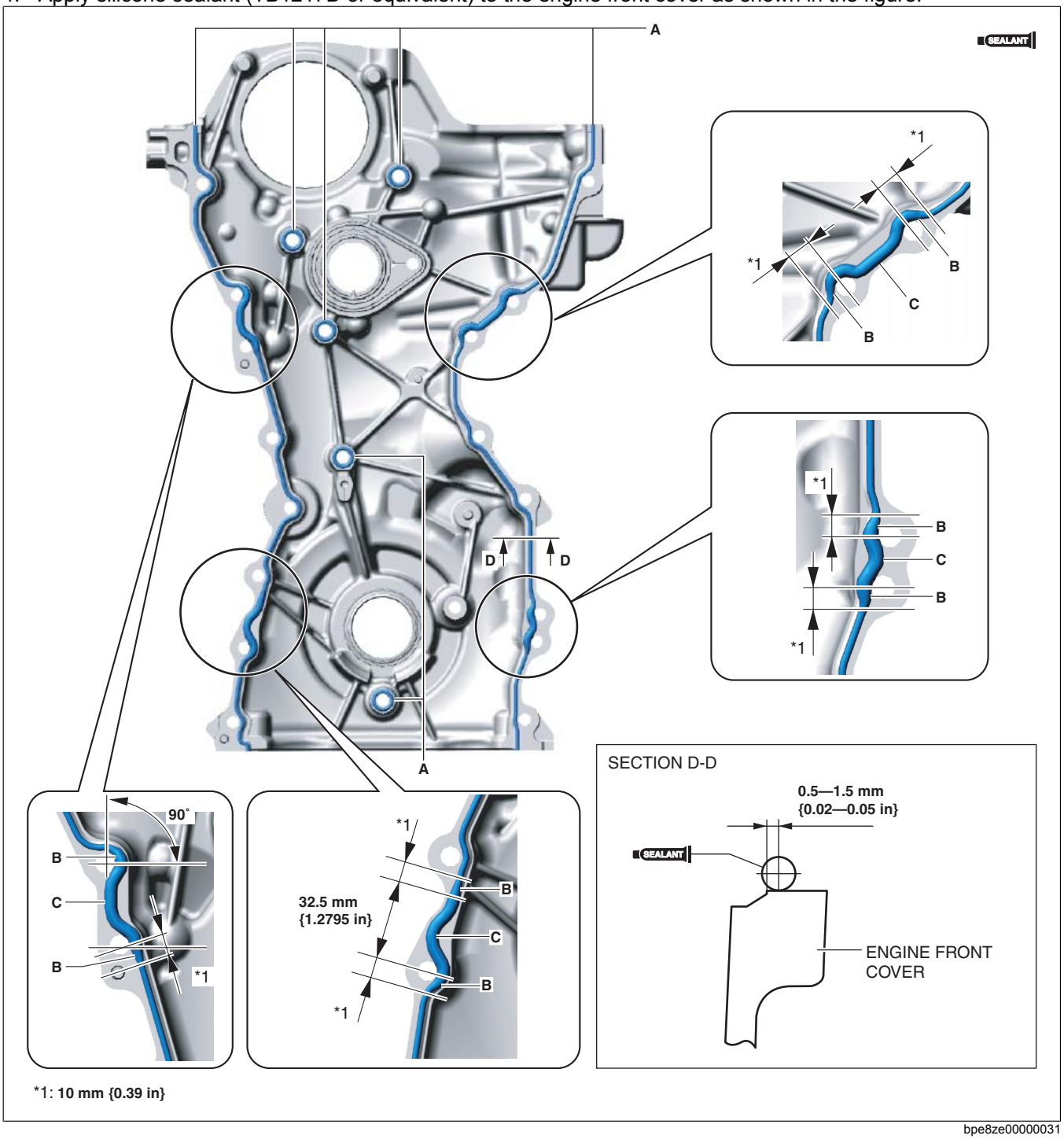
2. When reusing an engine front cover installation bolts, remove silicone sealant adhering to the bolts.

**Caution**
- **If oil, dirt and silicone sealant remains on the silicone sealant application area, the silicone sealant will not seal which will cause oil leakage.**

3. Completely clean and remove oil, dirt, silicone sealant or other foreign matter that may be adhering to the engine front cover, cylinder head, and cylinder block.

**Caution**
- **Apply the silicone sealant in a single, unbroken line.**
- **To prevent silicone sealant from hardening, adhere the engine front cover and the cylinder block firmly within 10 min. after applying silicone sealant. After adhering them, tighten the installation bolts immediately.**

4. Apply silicone sealant (TB1217D or equivalent) to the engine front cover as shown in the figure.

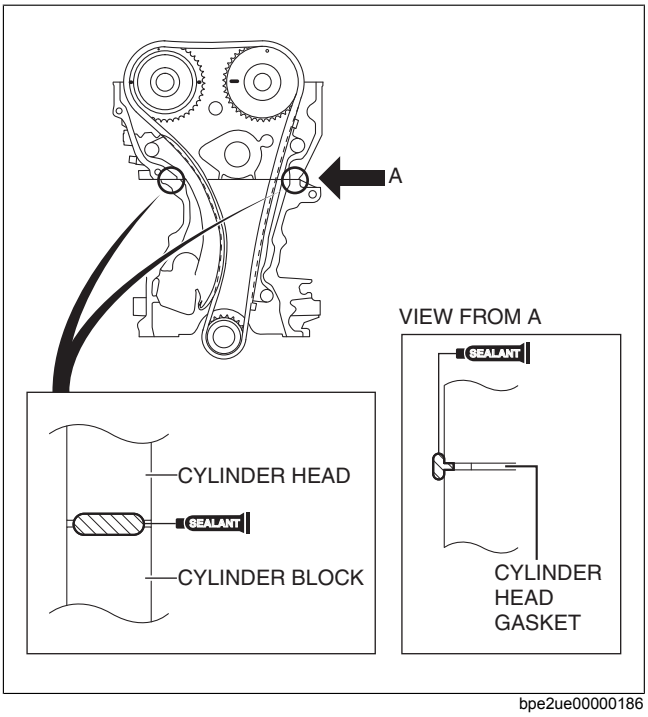



bpe8ze00000031

**Bead thickness**
**A: 2—6 mm {0.1—0.2 in}**
**B: 4—6 mm {0.16—0.23 in}**
**C: 4—8 mm {0.2—0.3 in}**

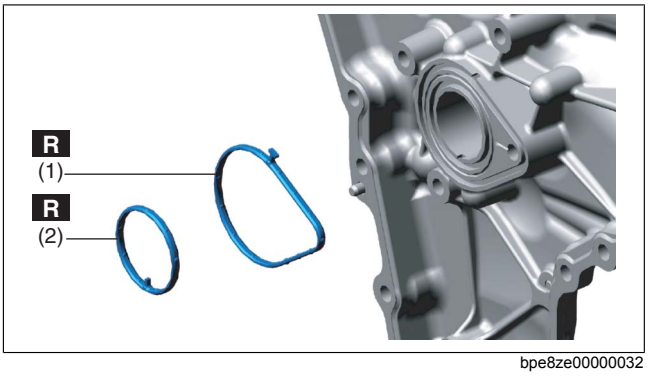
**Caution**
- **Apply the silicone sealant so that it goes into the cylinder head gasket.**

5. Apply silicone sealant (TB1217D or equivalent) to the areas shown in the figure.



bpe2ue00000186

**Caution**
- **Do not apply engine coolant to the O-ring (1). Otherwise, the O-ring could swell causing a seal malfunction.**
- **Do not apply oil (such as engine oil, ATF) to the O-ring (2). Otherwise, the O-ring could swell causing a seal malfunction.**



bpe8ze00000032

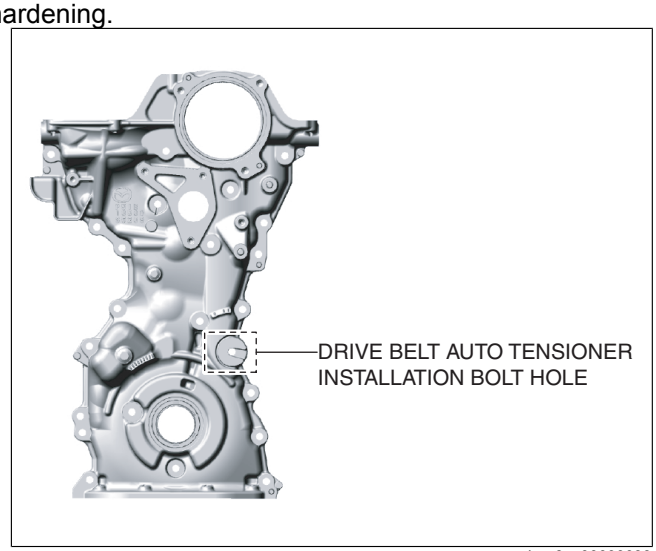
6. Install a new O-ring to the O-ring installation groove of the engine front cover.
7. Install the engine front cover to the engine.

**Note**
- Temporarily install an appropriate bolt to the drive belt auto tensioner installation bolt hole to prevent:
  - — A silicone sealant adhesion malfunction in the drive belt auto tensioner installation bolt hole.
  - — A bolt mis-installation due to silicone sealant hardening.

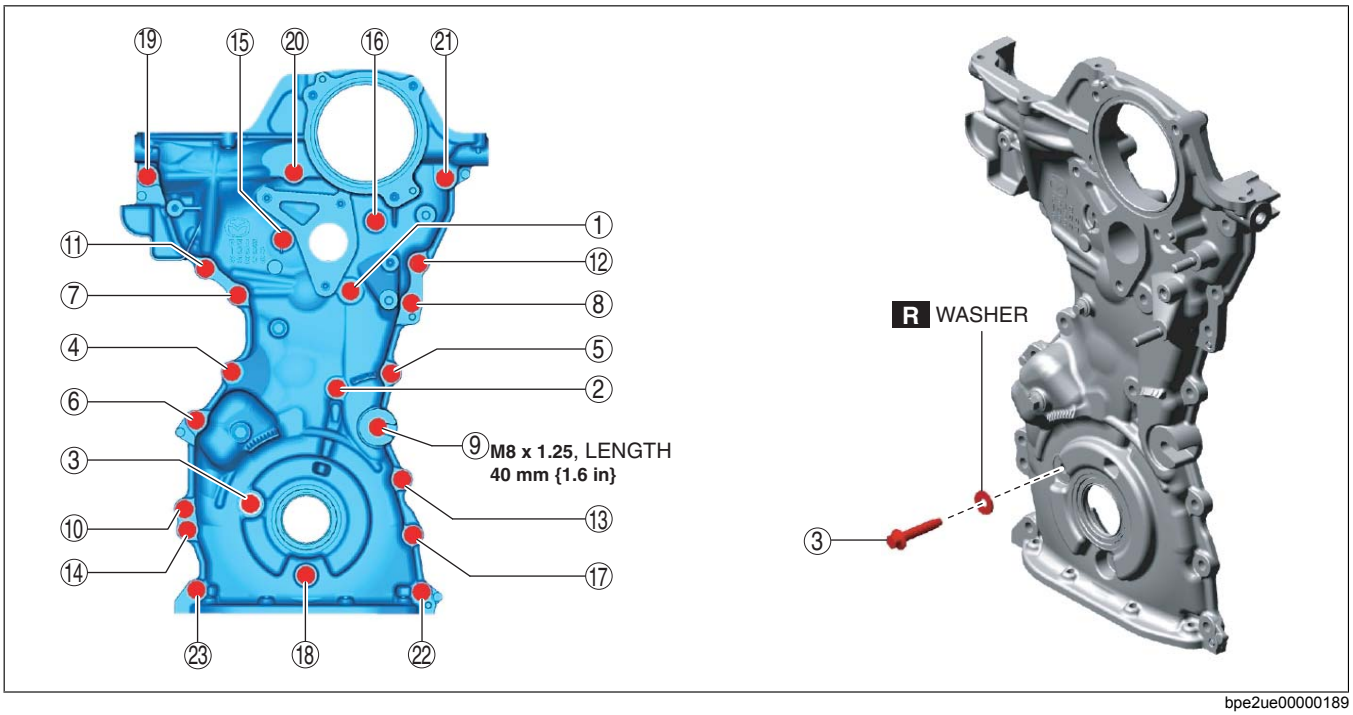
8. Prepare an appropriate **M8 × 1.25** bolt (length **40 mm {1.6 in}**).

**Caution**
- **For the number 3 bolt of the tightening order, install the bolt with new washer.**

9. Tighten the engine front cover installation bolts in the order shown in the figure.



bpe8ze00000033




bpe2ue00000189

**Tightening torque**
**20—26 N·m {2.1—2.6 kgf·m, 15—19 ft·lbf}**

10. Remove the bolt installed to the drive belt auto tensioner installation bolt hole when installing the drive belt auto tensioner.

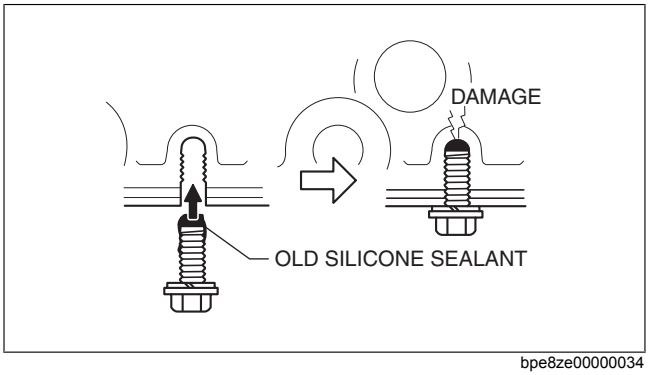
**Oil Pan Assembly Note (Except MX-5 (ND))**

1. Completely clean and remove any oil, dirt, sealant or other foreign matter that may be adhering to the cylinder block and oil pan.

**Caution**
- **Using bolts with the old silicone sealant adhering could cause cracks in the cylinder block, etc.**

2. When reusing the oil pan installation bolts, clean any old sealant from the bolts.

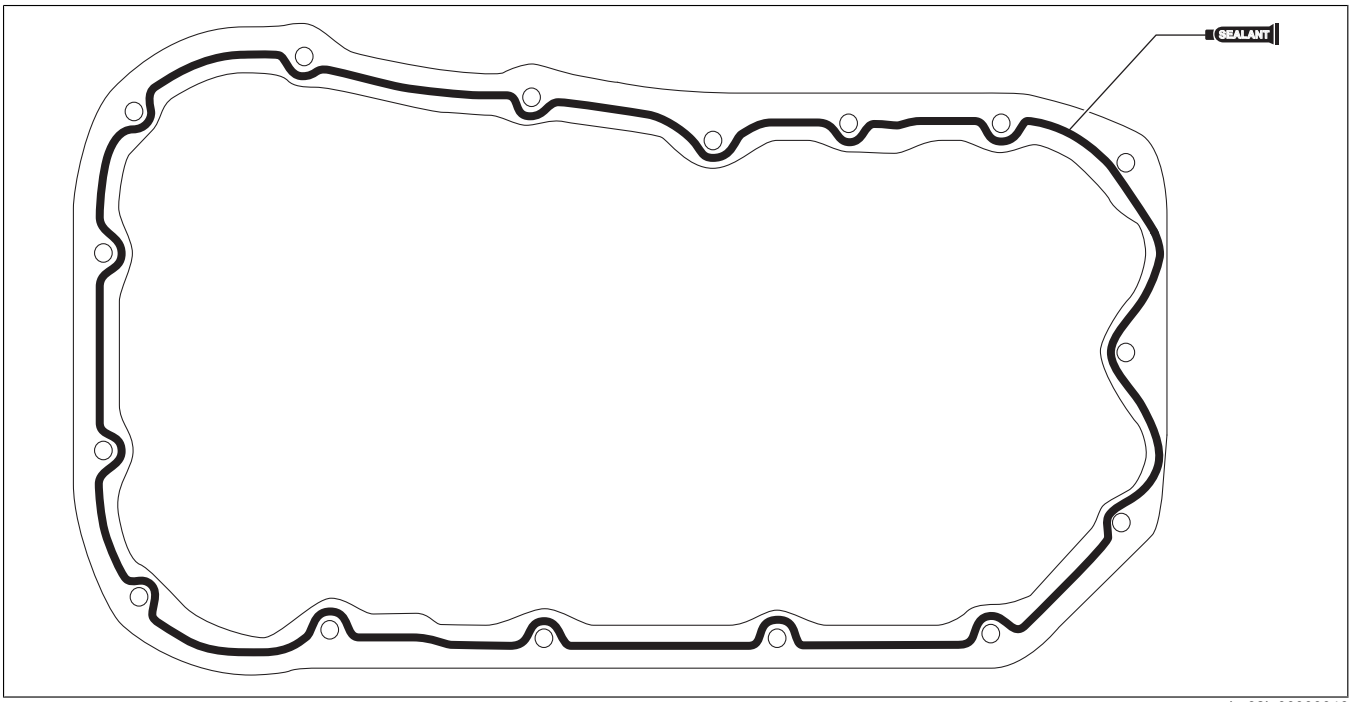
**Caution**
- **Apply the silicone sealant in a single, unbroken line around the whole perimeter.**
- **To prevent silicone sealant from hardening, adhere the oil pan to the cylinder block within 10 min. after silicone sealant is applied. Tighten the installation bolts completely soon after adhering.**




bpe8ze00000034

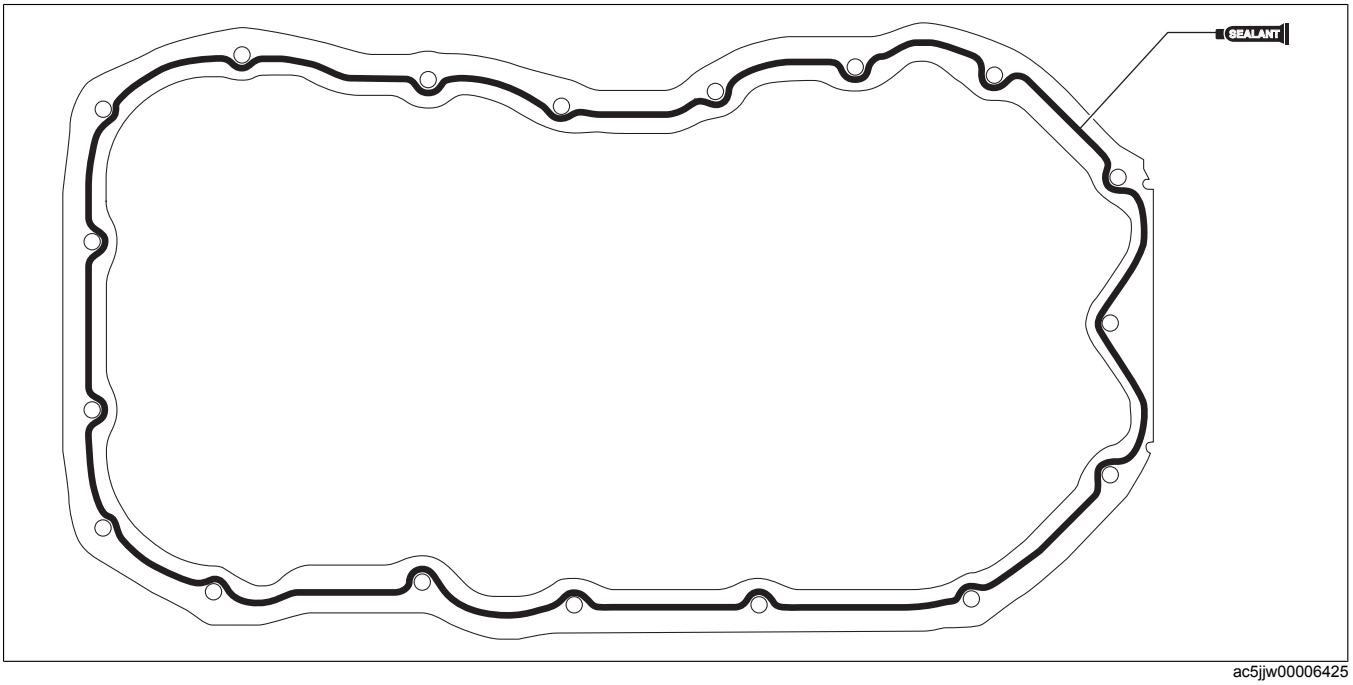
3. Apply silicone sealant (TB1217D or equivalent) to the oil pan along the inside of the bolt holes as shown in the figure.

**SKYACTIV-G 1.3, SKYACTIV-G 1.5**




bp32je00000046

**SKYACTIV-G 2.0**

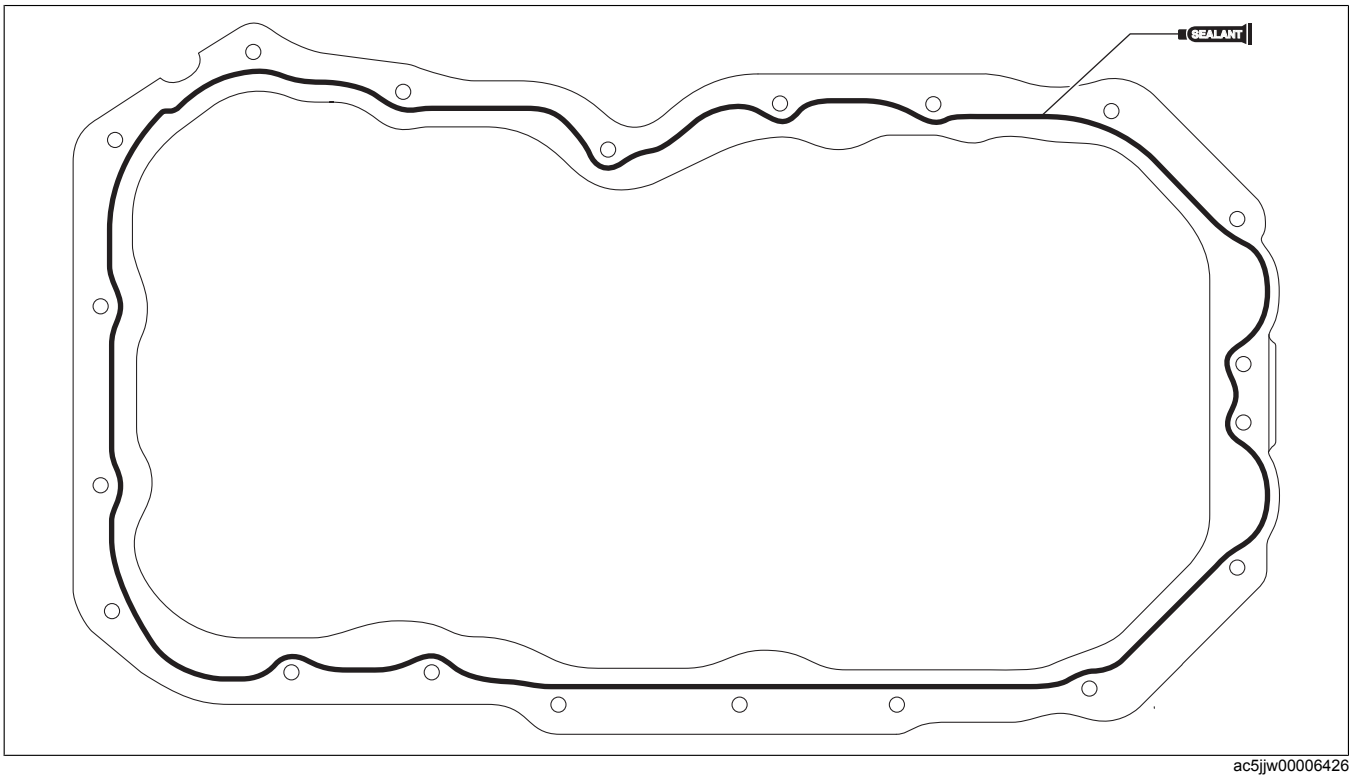



ac5jjw00006425

**Caution**

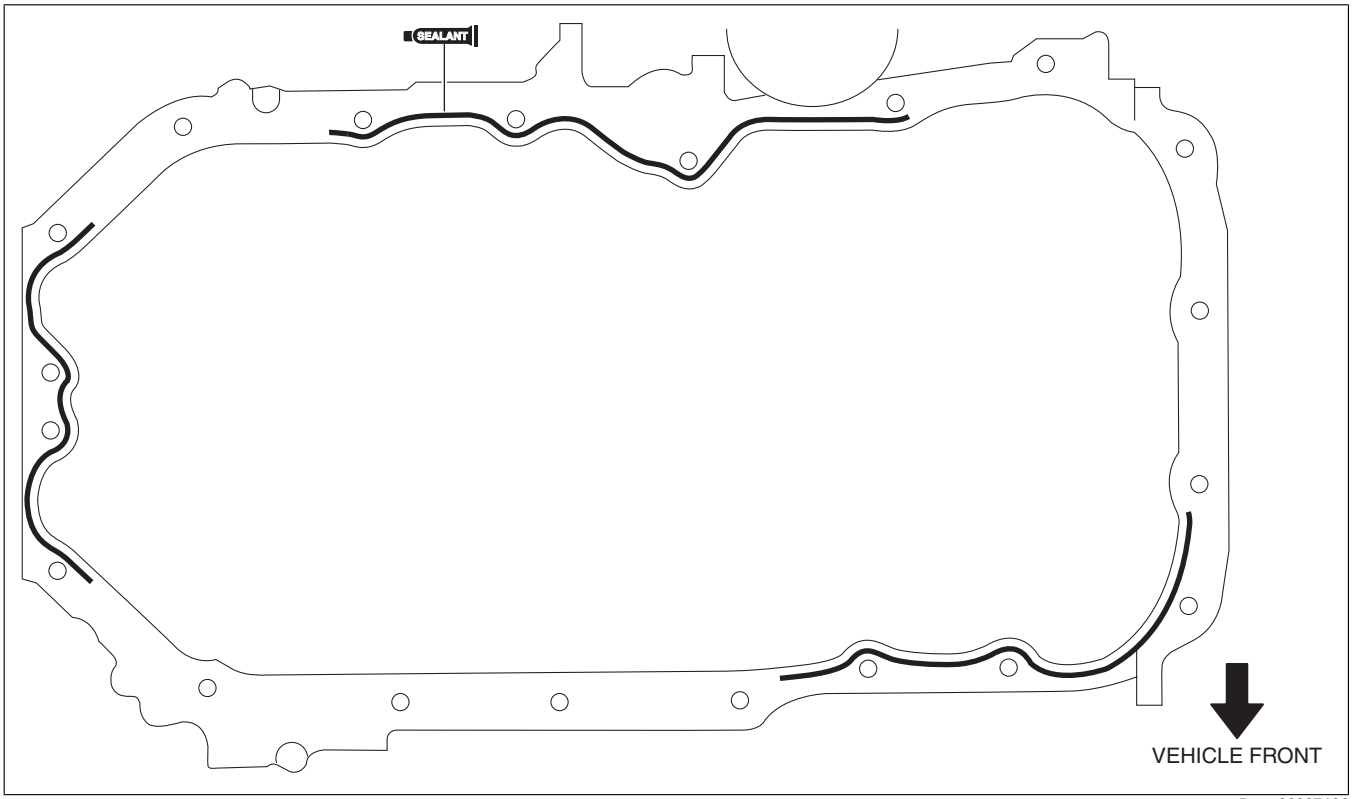
- **For SKYACTIV-G 2.5, also apply silicone sealant to the cylinder block.**

**SKYACTIV-G 2.5 oil pan**




ac5jjw00006426

**SKYACTIV-G 2.5 cylinder block**



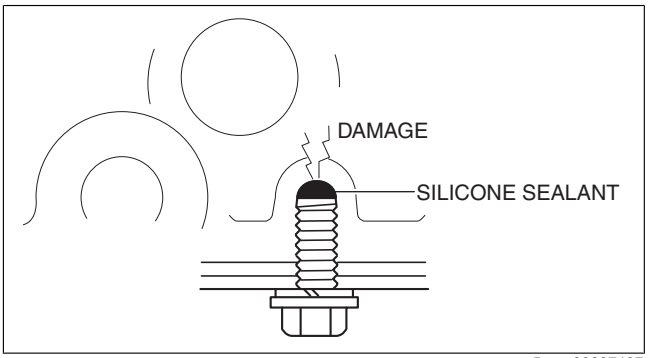

ac5wzw00007106

**Thickness**
**SKYACTIV-G 1.3, SKYACTIV-G 1.5, SKYACTIV-G 2.0: 2.0—6.0 mm {0.08—0.23 in}**
**SKYACTIV-G 2.5: 3.0—7.0 mm {0.12—0.27 in}**

4. Install the oil pan using the following procedure:

**Caution**

- **Remove any silicone sealant which oozes out into an oil pan installation bolt hole. If silicone sealant penetrates an oil pan installation bolt hole, the cylinder block could become damaged.**

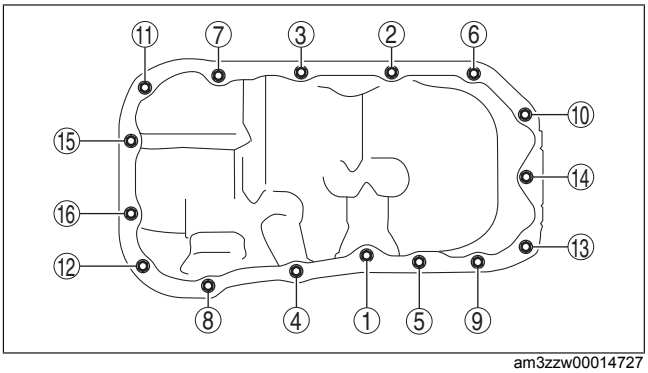


ac5wzw00007107

**SKYACTIV-G 1.3, SKYACTIV-G 1.5, SKYACTIV-G 2.0**

1. Install the oil pan to the cylinder block.
2. Tighten the bolts in the order shown in the figure.

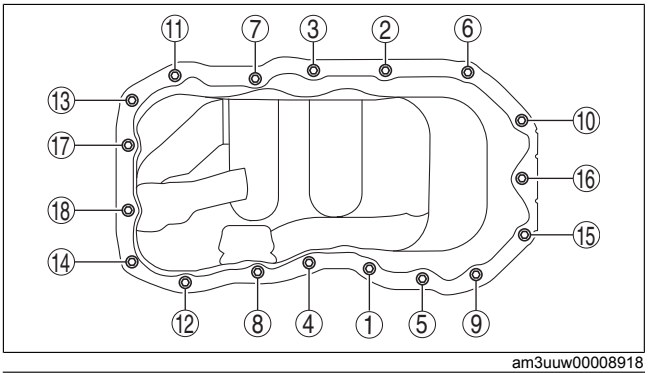
**SKYACTIV-G 1.3, SKYACTIV-G 1.5**



am3zzw00014727

**SKYACTIV-G 2.0**

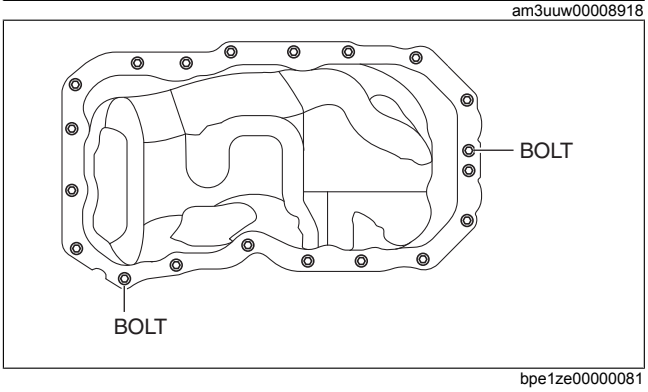
**Tightening torque**
**8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}**



am3uuw00008918

**SKYACTIV-G 2.5**

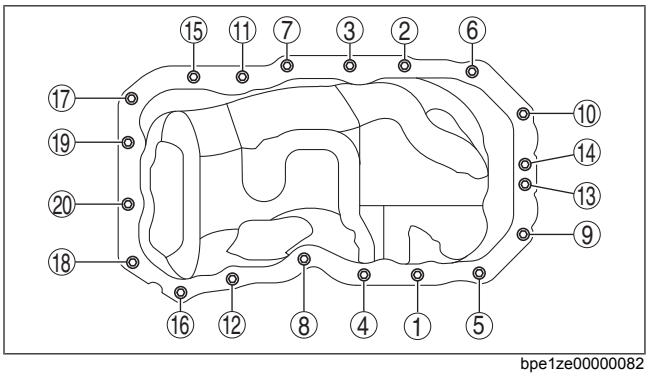
1. Install the oil pan to the cylinder block.
2. Temporarily tighten the two bolts shown in the figure.



bpe1ze00000081

3. Tighten the bolts in the order shown in the figure.

**Tightening torque**
**8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}**



bpe1ze00000082

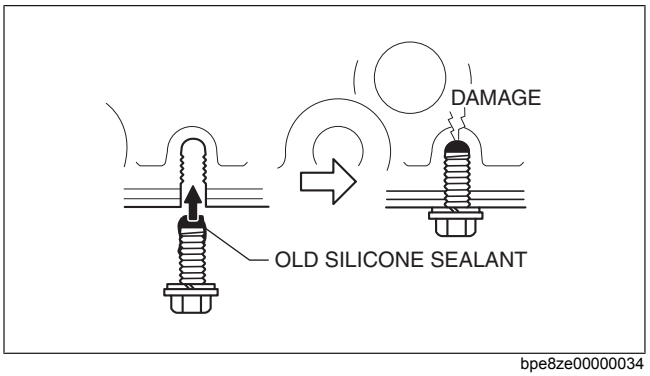
**Oil Pan Assembly Note (MX-5 (ND))**

1. Completely clean and remove any oil, dirt, silicone sealant or other foreign matter that may be adhering to the cylinder block and oil pan.

**Caution**
- **Using bolts with the old silicone sealant adhering could cause cracks in the cylinder block.**

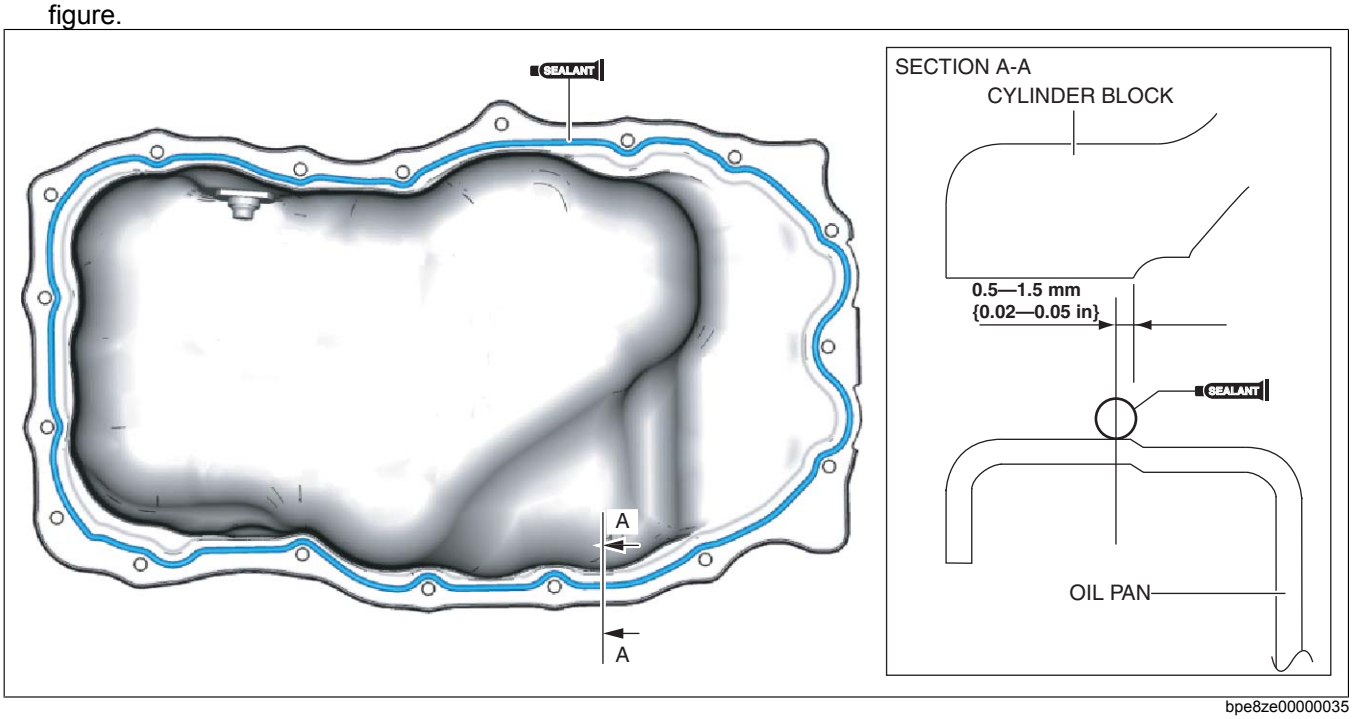
2. When reusing the oil pan installation bolts, clean any old silicone sealant from the bolts.

**Caution**
- **Apply the silicone sealant in a single, unbroken line around the whole perimeter.**
- **To prevent silicone sealant from hardening, adhere the oil pan to the cylinder block within 10 min after silicone sealant is applied. Tighten the installation bolts completely soon after adhering.**



bpe8ze00000034

3. Apply silicone sealant (TB1217D or equivalent) to the oil pan along the inside of the bolt holes as shown in the figure.

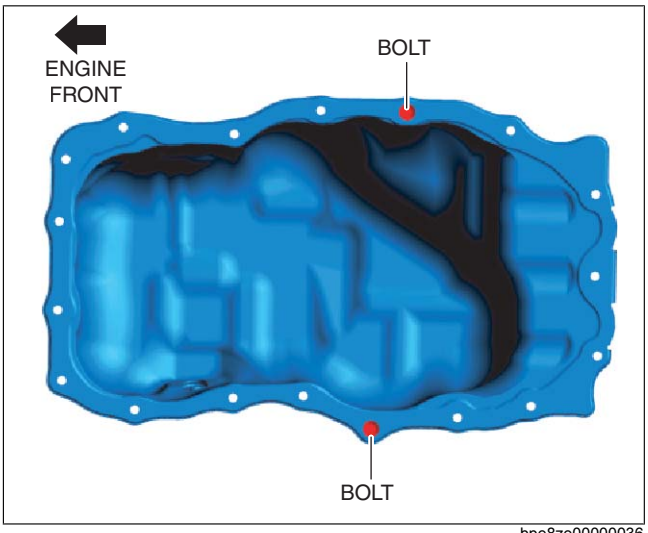


bpe8ze00000035

**Thickness**
**3.0—7.0 mm {0.12—0.27 in}**

4. Install the oil pan using the following procedure:
(1) Install the oil pan to the cylinder block.

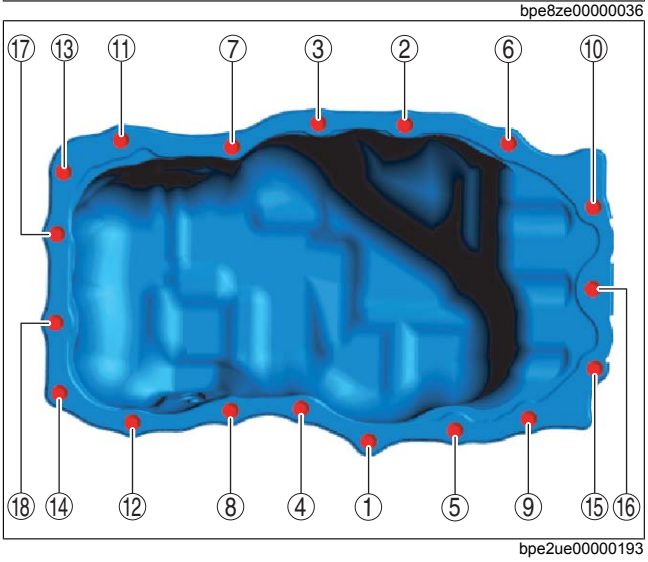
(2) Temporarily tighten the two bolts shown in the figure.



bpe8ze00000036

(3) Tighten the bolts in the order shown in the figure.

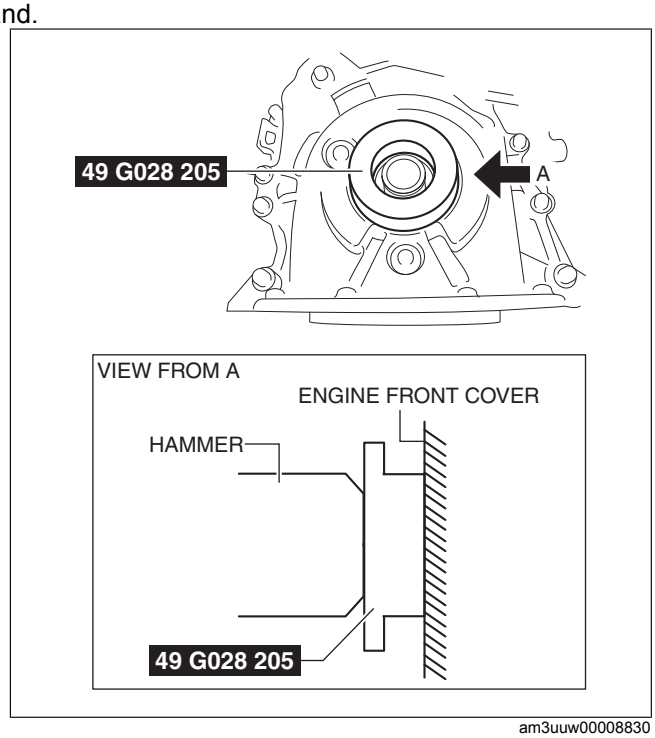
**Tightening torque**
**8—11 N·m {82—112 kgf·cm, 71—97 in·lbf}**



bpe2ue00000193

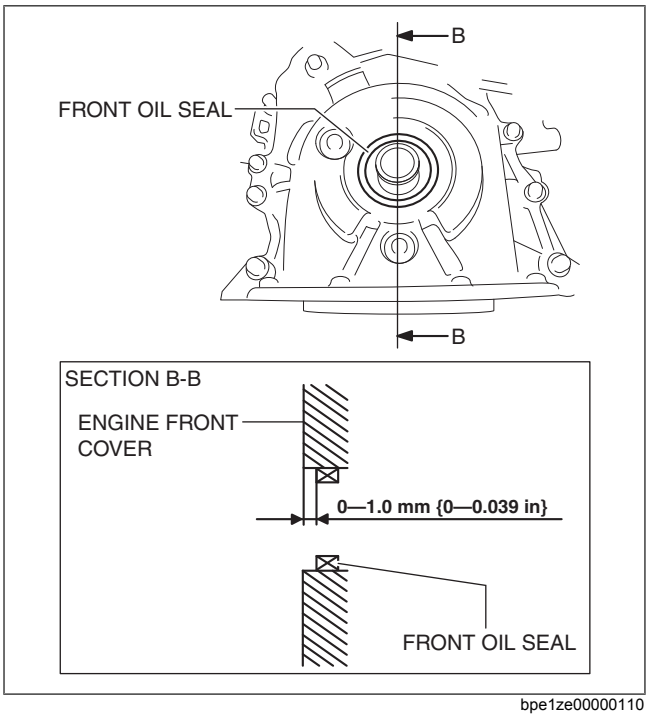
**Front Oil Seal Assembly Note**

1. Apply clean engine oil to the inner surface of a new front oil seal.
2. Insert the front oil seal into the engine front cover by hand.
3. Tap the front oil seal in evenly using the **SST** and a hammer.



am3uuw00008830

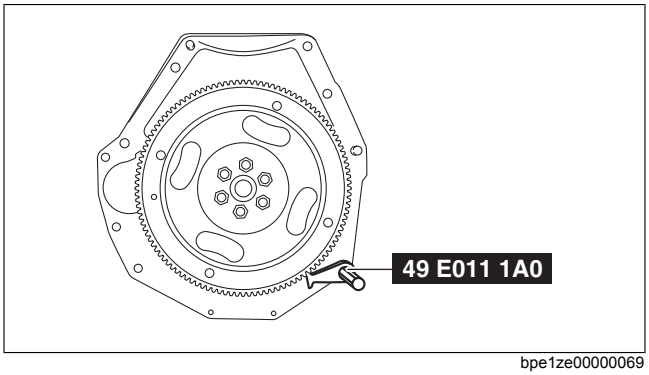
**Front oil seal press on amount**
**0—1.0 mm {0—0.039 in} from edge surface of engine front cover**



bpe1ze00000110

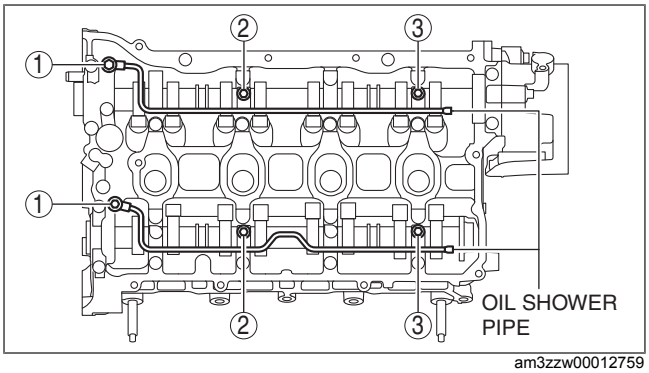
**Crankshaft Pulley Lock Bolt Assembly Note**
1. Hold the crankshaft using the **SST**.
2. Tighten the crankshaft pulley lock bolt in the order shown in the following two steps.

**Tightening procedure**
**Step 1: 90—110 N·m {9.2—11 kgf·m, 67—81 ft·lbf}**
**Step 2: 55—65°**



bpe1ze00000069

**Oil Shower Pipe Assembly Note**
1. Install the oil shower pipe in the order shown in the figure.



am3zzw00012759

**Tightening torque**

| Installation position | Tightening torque |
|---|---|
| 1 | 9—12 N·m {92—122 kgf·cm, 80—106 in·lbf} |
| 2, 3 | 8.0—9.0 N·m {82—91 kgf·cm, 71—79 in·lbf} |

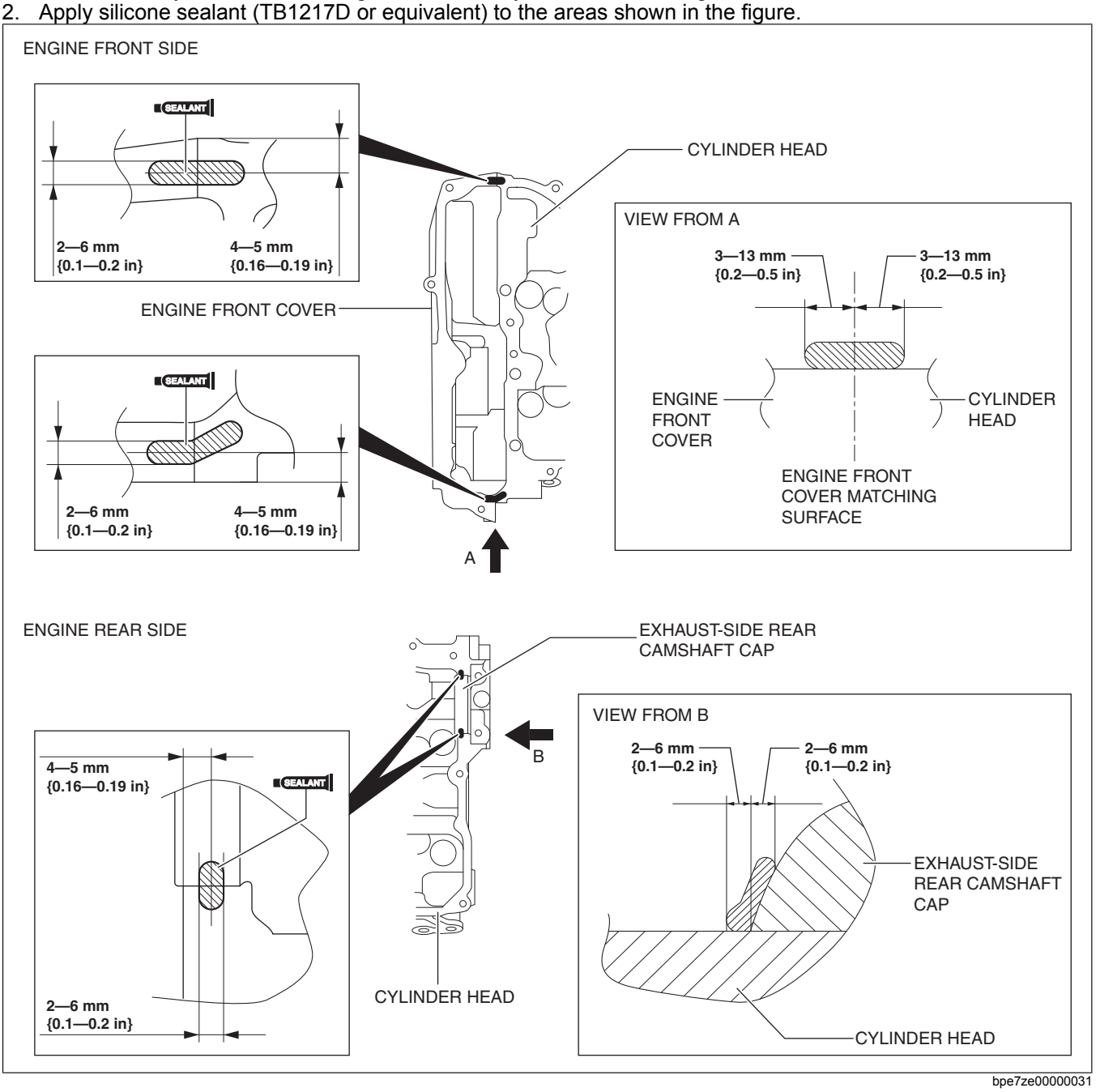
**Cylinder Head Cover Assembly Note (Except MX-5 (ND))**

**Caution**
- **To assure the sealing performance of the cylinder head cover, be careful of the following:**

— **Verify that the cylinder head cover gasket is inserted into the cylinder head cover groove and install the cylinder head cover.**
— **Completely clean and remove any oil, dirt, sealant or other foreign material from the seal surface.**

- **To prevent silicone sealant from hardening, adhere the cylinder head cover and the cylinder head within 10 min. after silicone sealant is applied. Tighten the installation bolts completely soon after adhering.**

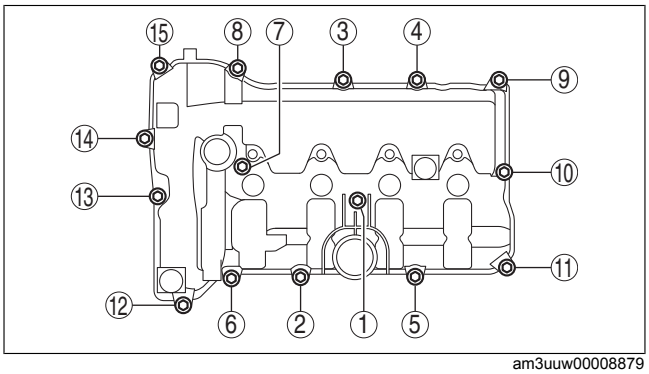
1. Insert a new cylinder head cover gasket into the cylinder head cover groove.
2. Apply silicone sealant (TB1217D or equivalent) to the areas shown in the figure.




bpe7ze00000031

3. Install the cylinder head cover.
4. Temporarily tighten the cylinder head cover bolts.
5. Tighten the cylinder head cover bolts in the order shown in the figure.

**With electric variable valve timing**

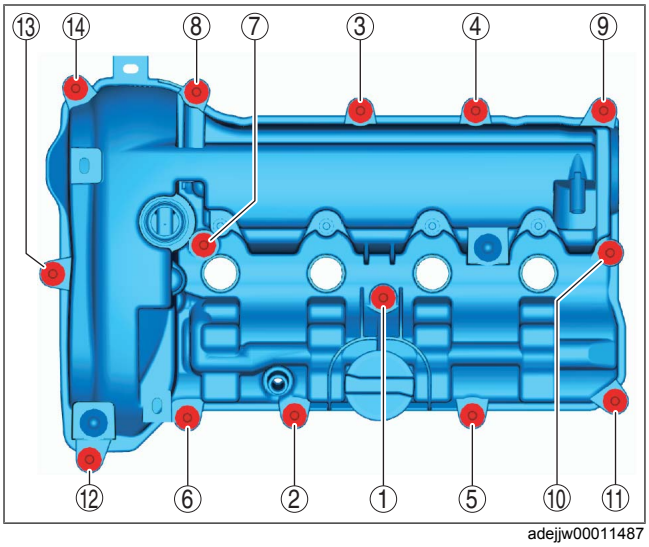


am3uuw00008879

**Without electric variable valve timing**

**Tightening torque**
**6.0—7.0 N·m {62—71 kgf·cm, 54—61 in·lbf}**

6. Measure the tightening torque again and verify that it is **6.0 N·m {61 kgf·cm, 53 in·lbf}** or more.

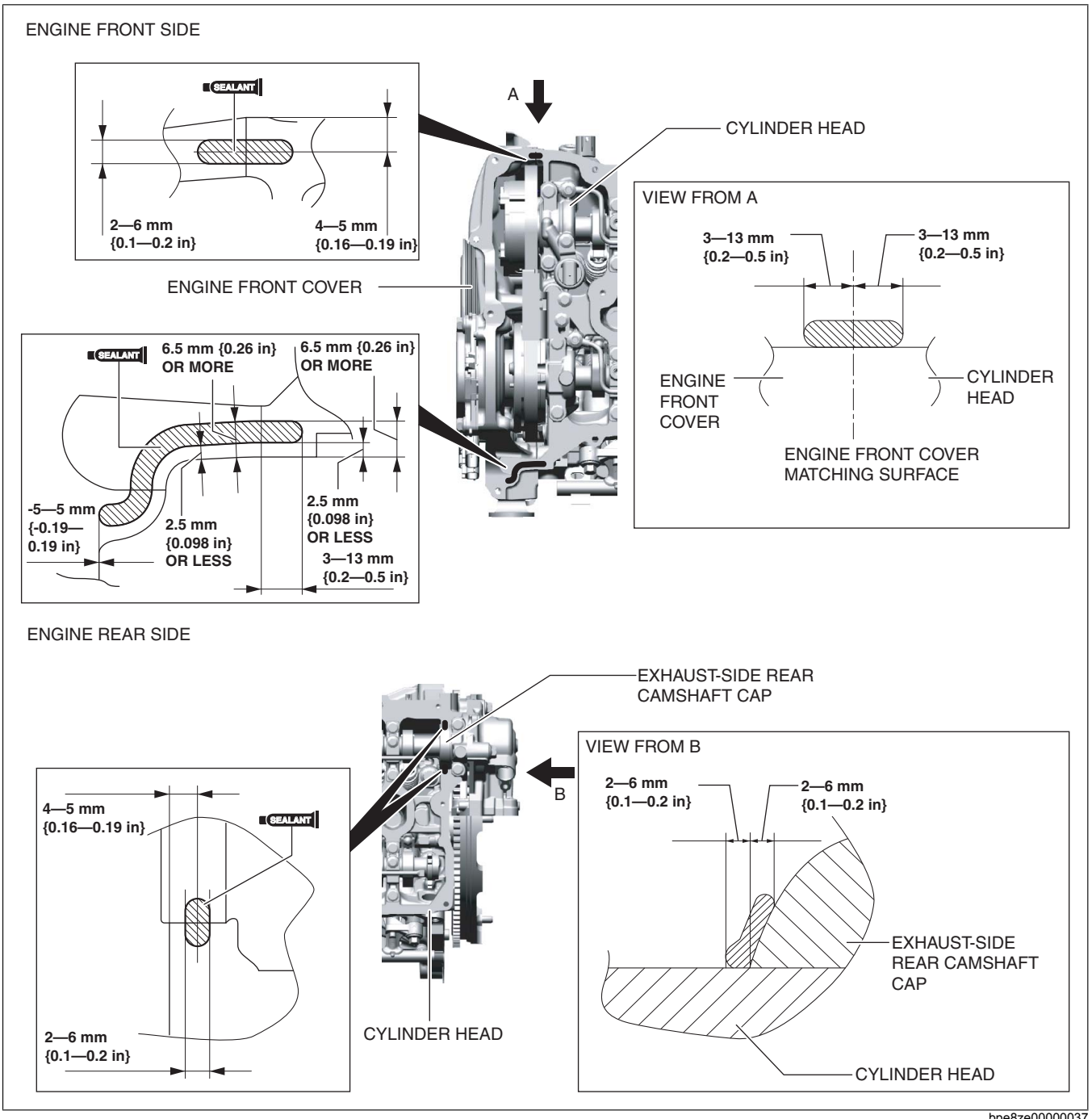


adejjw00011487

**Cylinder Head Cover Installation Note (MX-5 (ND))**

**Caution**
- **To assure the sealing performance of the cylinder head cover, be careful of the following:**
  - **— Verify that the cylinder head cover gasket is inserted into the cylinder head cover groove and install the cylinder head cover.**
  - **— Completely clean and remove oil, dirt, silicone sealant or other foreign matter from the seal surface.**
- **To prevent silicone sealant from hardening, adhere the cylinder head cover and the cylinder head firmly within 10 min. after applying silicone sealant. After adhering them, tighten the installation bolts immediately.**

1. Insert a new cylinder head cover gasket into the cylinder head cover groove.
2. Apply silicone sealant (TB1217D or equivalent) to the areas shown in the figure.

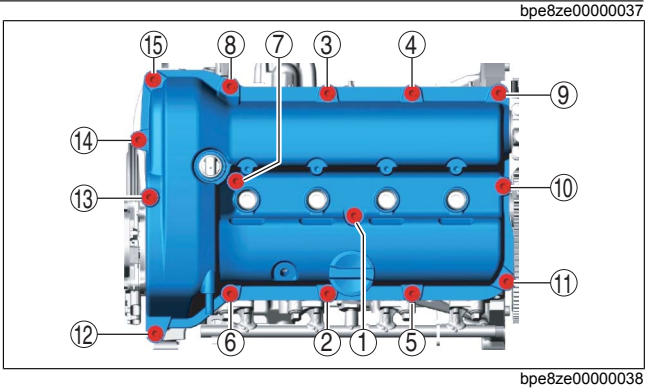



bpe8ze00000037

3. Tighten the cylinder head cover bolts in the order shown in the following 3 steps.

   **Tightening torque**
   **Step 1: 6—10 N·m {62—101 kgf·cm, 54—88 in·lbf}**
   **Step 2: 6—10 N·m {62—101 kgf·cm, 54—88 in·lbf}**
   **Step 3: 6—10 N·m {62—101 kgf·cm, 54—88 in·lbf}**

4. Measure the tightening torque again and verify that it is **6.0 N·m {61 kgf·cm, 53 in·lbf}** or more.




bpe8ze00000038